

Core Web Vitals Performance Analysis

CSE211 - (Web Programming)

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Group 5:

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Executive Summary

The report provides a very specific and detailed analysis of the Core Web Vital performance by using 3 different domain websites: Nike as an e-commerce platform, Al-Jazeera as a news/media outlet, and KhanAcademy as an educational institution. The analysis used in finding and evaluating each website's key performance metrics both globally and locally, such as LCP, INP, and CLS on Mobile and Desktop, were adapted from Google Pagespeed, which includes DevTools.

These findings (that will be discussed later on) indicate that desktop outperformed mobiles in all of the metrics measured, while mobile really needs improvement with its LCP and CLS. One example that will be shown later on is Nike's LCP (Mobile), where it got 3.5 seconds and its CLS-0.182, which means it needs to be improved. Another example is Al-Jazeera LCP (mobile), which got 4.6s (poor). Common reasons or issues which affect performance: for example, large image sizes, render blocking JS/CSS, and slower mobile CPUs speed. Lighthouse has also highlighted potential optimizations like reducing JS, improving image delivery, and implementing efficient caching.

Methodology

The analyses discovered throughout this report employed a unique, structured 3-Phase approach to evaluate and compare the Core Web Vitals across 3 domains.

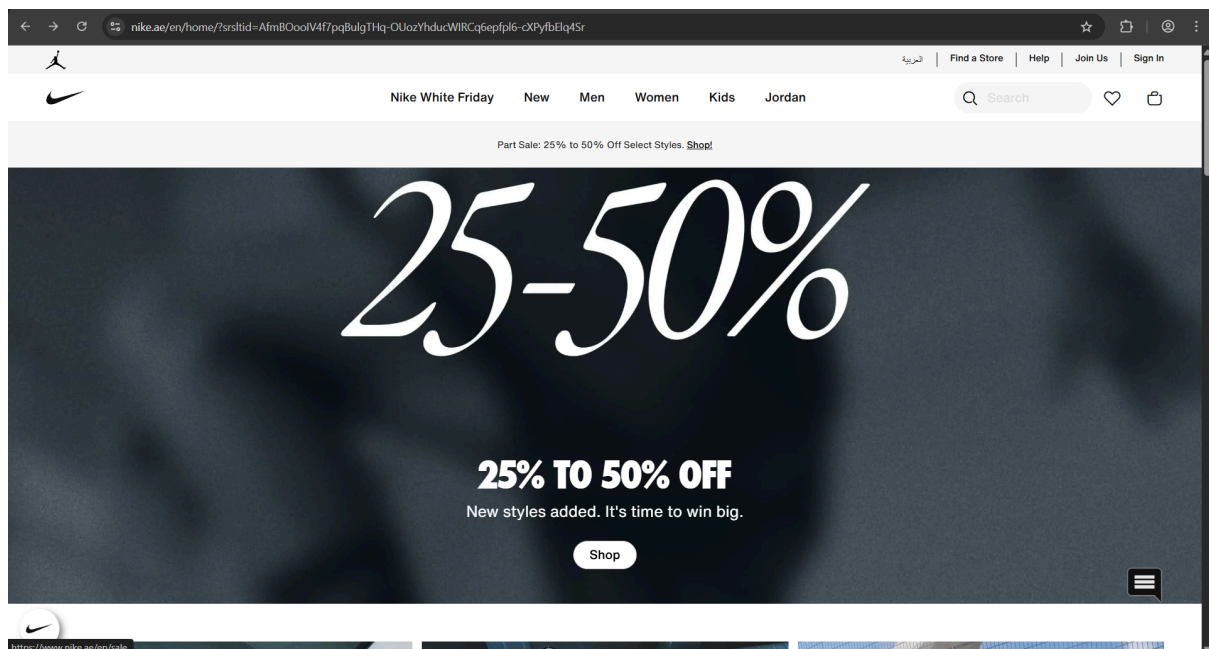
Phase 1 involves us selecting the representor website from each category (e-commerce, news/media, and education) and then searching through DevTools and finding their technology stacks and server infrastructure. Phase 2 consists of recording both Global and Local performance data. Both Local and Global will be recorded using Google PageSpeed Insights, which includes DevTools. Global performance data comes from the 'Chrome User Experience Report' over a period of 28 days from real-world users. Phase 3 focuses on comparing all the analysis gathered from the 3 websites and identifying the worst performer, diagnosing its bottlenecks, and then creating ways and how we could optimize and address those critical failures.

1. Website Selection & Technology Analysis

The 3 Websites that will be tested upon:

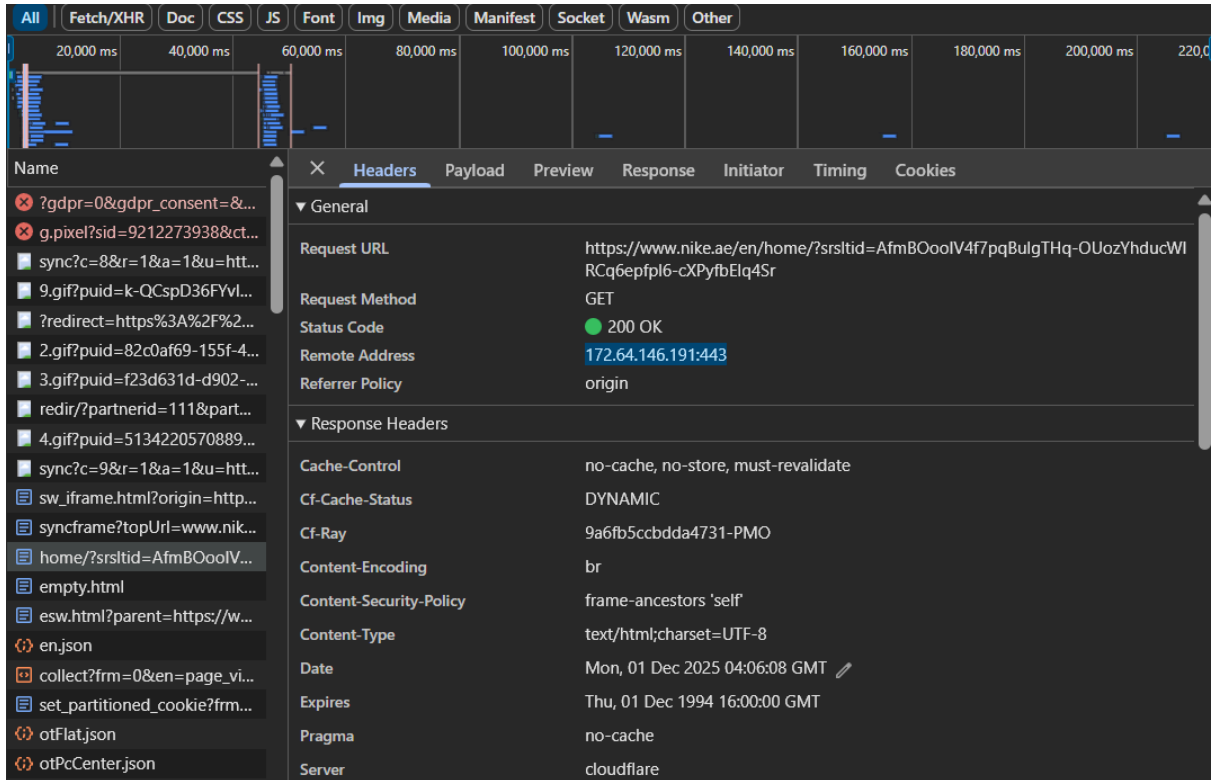
Domain Types	Website Name	URL
1. E-commerce Platform	Nike (UAE/.ae)	https://www.nike.ae/en/home/
2. News/Media Outlet	Al-Jazeera	https://www.aljazeera.com/
3. Educational Institution	Khan Academy	https://www.khanacademy.org/

1.1 Nike (E-commerce Platform)



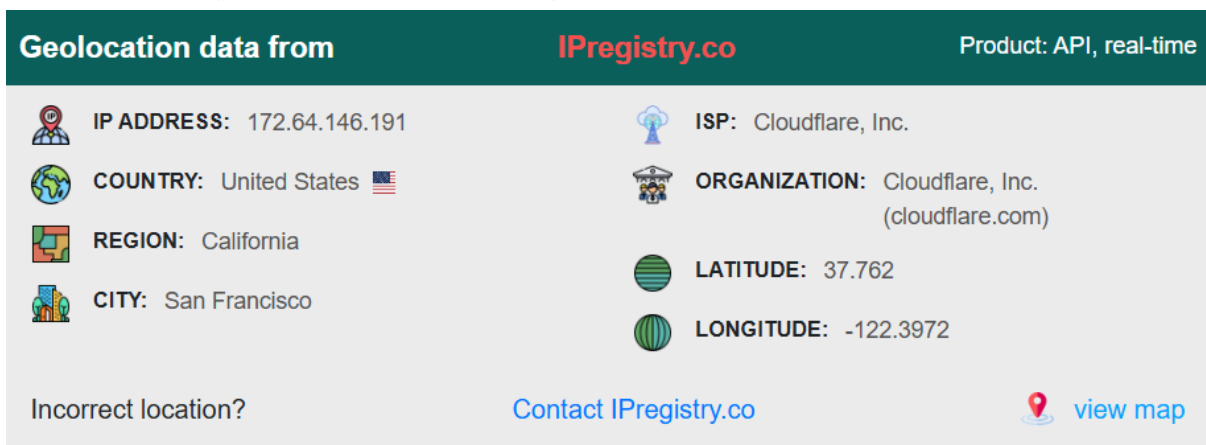
URL - (<https://www.nike.ae/en/home/>)

1.11 Server Location



The screenshot shows the Chrome DevTools Network tab with the 'Headers' panel selected. The request is a GET to `https://www.nike.ae/en/home/?srsid=AfmBOooIV4f7pqBulgTHq-OUozYhducWIRcQ6epfpl6-cXPyfbElq4Sr`. The status is 200 OK. The Remote Address is `172.64.146.191:443`. The Response Headers include: Cache-Control: no-cache, no-store, must-revalidate; Cf-Cache-Status: DYNAMIC; Cf-Ray: 9a6fb5ccbdda4731-PMO; Content-Encoding: br; Content-Security-Policy: frame-ancestors 'self'; Content-Type: text/html; charset=UTF-8; Date: Mon, 01 Dec 2025 04:06:08 GMT; Expires: Thu, 01 Dec 1994 16:00:00 GMT; Pragma: no-cache; Server: cloudflare.

Extra - (to find server location)

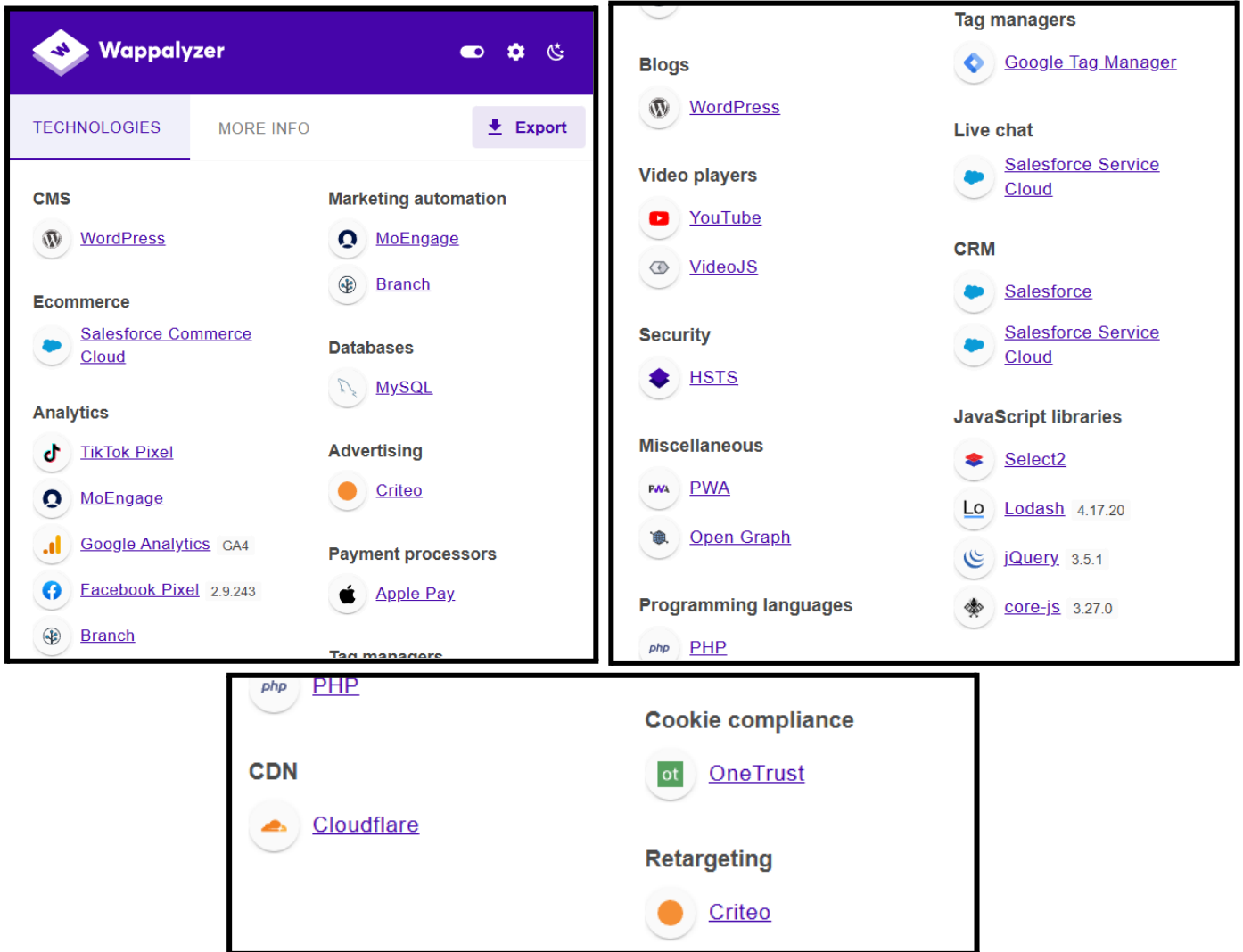


The screenshot shows the IPRegistry.co geolocation data for IP 172.64.146.191. The data includes: IP ADDRESS: 172.64.146.191; COUNTRY: United States; REGION: California; CITY: San Francisco; ISP: Cloudflare, Inc.; ORGANIZATION: Cloudflare, Inc. (cloudflare.com); LATITUDE: 37.762; LONGITUDE: -122.3972. There is a link to 'view map' and a 'Contact IPRegistry.co' link.

Website Name	Organization hosting the Server	Server IP Address	Server Country	Server Region/City
Nike	Cloudflare	172.64.146.191 (Port 443)	USA	California / San-Francisco

1.12 Technology stack identification (using Both Wappalyzer and DevTools)

1.121 Wappalyzer:



The screenshot displays the Wappalyzer interface, which categorizes detected technologies into several groups:

- TECHNOLOGIES** (Main View):
 - CMS**: WordPress
 - Ecommerce**: Salesforce Commerce Cloud
 - Analytics**: TikTok Pixel, MoEngage, Google Analytics GA4, Facebook Pixel 2.9.243, Branch
 - Marketing automation**: MoEngage, Branch
 - Databases**: MySQL
 - Advertising**: Criteo
 - Payment processors**: Apple Pay
 - Tag managers**: (partially visible)
- Tag managers**: Google Tag Manager
- Live chat**: Salesforce Service Cloud
- CRM**: Salesforce, Salesforce Service Cloud
- JavaScript libraries**: Select2, Lodash 4.17.20, jQuery 3.5.1, core-js 3.27.0
- Miscellaneous**: PWA, Open Graph
- Programming languages**: PHP
- CDN**: Cloudflare
- Cookie compliance**: OneTrust
- Retargeting**: Criteo

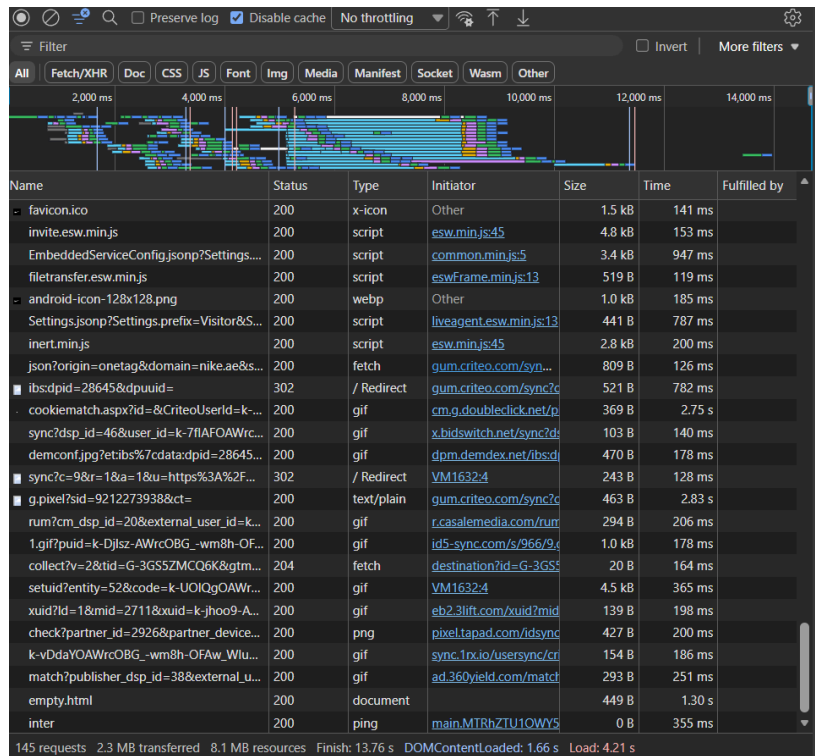
Overview:

Content Management System (CMS)	WordPress - (for blog/marketing) & The rest of the website uses Salesforce Commerce Cloud
Frontend Frameworks	None (Uses Custom JS & jQuery)

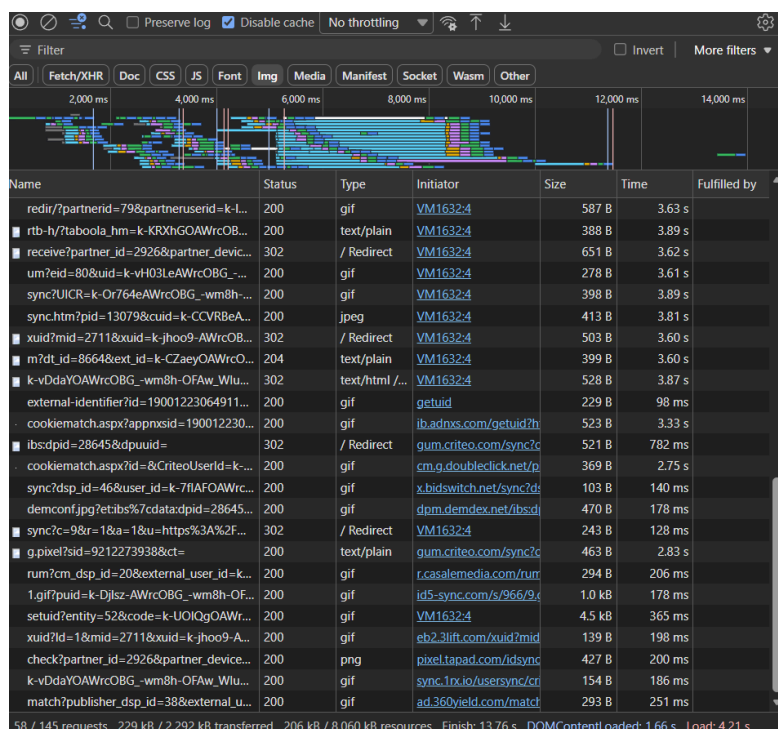
1.13 Measurements:

1.131 Measurement 1 (Morning 8 - 9 A.M.):

Page Load Summary →

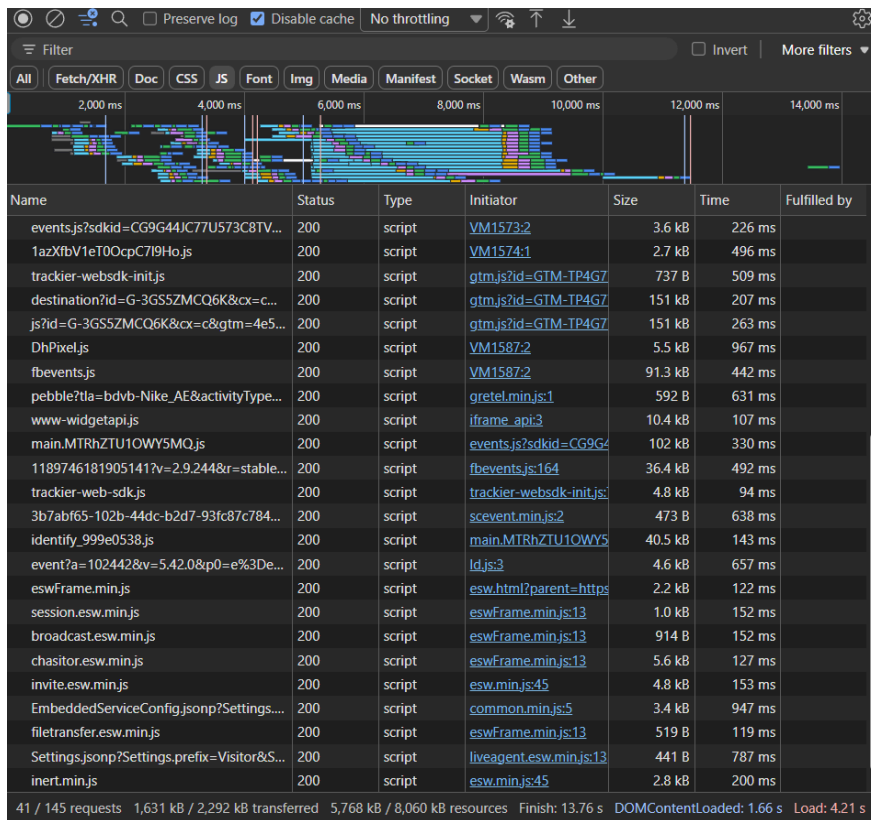


145 requests 2.3 MB transferred 8.1 MB resources Finish: 13.76 s DOMContentLoaded: 1.66 s Load: 4.21 s



← Image Load Impact

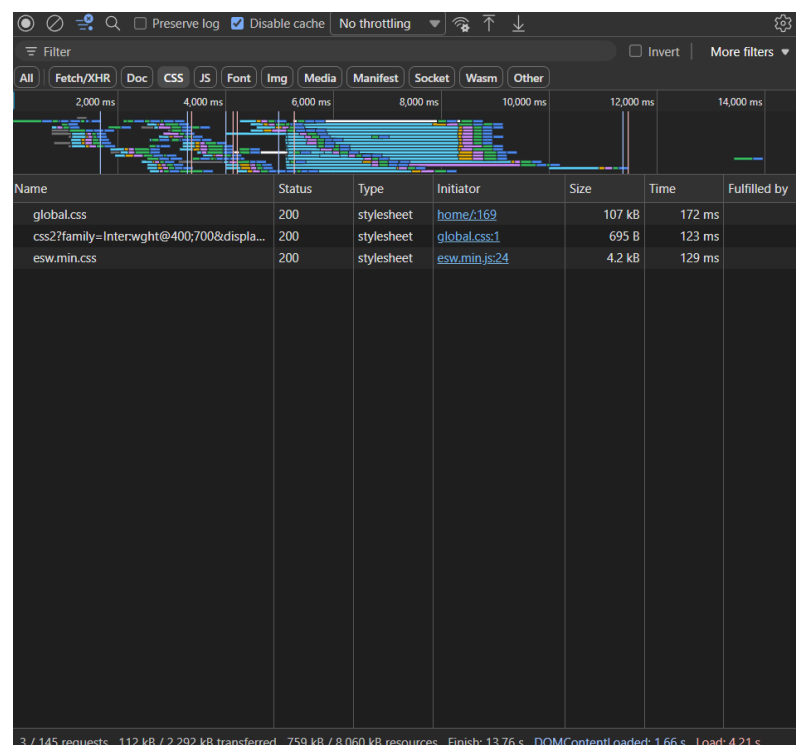
58 / 145 requests 229 kB / 2,292 kB transferred 206 kB / 8,060 kB resources Finish: 13.76 s DOMContentLoaded: 1.66 s Load: 4.21 s



← JS Load Impact

41 / 145 requests 1,631 kB / 2,292 kB transferred 5,768 kB / 8,060 kB resources Finish: 13.76 s DOMContentLoaded: 1.66 s Load: 4.21 s

CSS Load
Impact →



3 / 145 requests 112 kB / 2,292 kB transferred 759 kB / 8,060 kB resources Finish: 13.76 s DOMContentLoaded: 1.66 s Load: 4.21 s

1.1311 MSRMNT-1 Table:

Metric	Value
Total HTTP Requests	145
Total Page Size	2.3 MB (transferred)
DOMContentLoaded	1.66s
Load Time	4.21s
Finish Time	13.76s
Total CSS Requests + (size)	3-req (112KB transferred)
Total JS Requests + (size)	41-req (1,631KB transferred)
Total Image Requests + (size)	58-req (229KB transferred)

1.132 Measurement 2 (Afternoon 5 - 6 P.M):

Page Load Summary



139 requests 2.0 MB transferred 7.5 MB resources Finish: 9.89 s DOMContentLoaded: 1.65 s Load: 3.73 s

Image Load Impact



55 / 139 requests 134 kB / 2,037 kB transferred 119 kB / 7,527 kB resources Finish: 9.89 s DOMContentLoaded: 1.65 s Load: 3.73 s

JS Load Impact



40 / 139 requests 1,473 kB / 2,037 kB transferred 5,321 kB / 7,527 kB resources Finish: 9.89 s DOMContentLoaded: 1.65 s Load: 3.73 s

CSS Load Impact



3 / 139 requests 112 kB / 2,037 kB transferred 759 kB / 7,527 kB resources Finish: 9.89 s DOMContentLoaded: 1.65 s Load: 3.73 s

1.1321 MSRMNT-2 Table:

Metric	Value
Total HTTP Requests	139
Total Page Size	2.0 MB (transferred)
DOMContentLoaded	1.65s
Load Time	3.73s
Finish Time	9.89s
Total CSS Requests + (size)	3-req (112KB transferred)
Total JS Requests + (size)	40-req (1,473KB transferred)
Total Image Requests + (size)	55-req (134KB transferred)

1.133 Measurement 3 (Night 11 - 12 P.M):

Page Load Summary



142 requests 2.4 MB transferred 7.8 MB resources Finish: 8.44 s DOMContentLoaded: 1.93 s Load: 4.53 s

Image Load Impact



59 / 142 requests 332 kB / 2,361 kB transferred 310 kB / 7,801 kB resources Finish: 8.44 s DOMContentLoaded: 1.93 s Load: 4.53 s

JS Load Impact



41 / 142 requests 1,631 kB / 2,361 kB transferred 5,768 kB / 7,801 kB resources Finish: 8.44 s DOMContentLoaded: 1.93 s Load: 4.53 s

CSS Load Impact



3 / 142 requests 112 kB / 2,361 kB transferred 759 kB / 7,801 kB resources Finish: 8.44 s DOMContentLoaded: 1.93 s Load: 4.53 s

1.1331 MSRMNT-3 Table:

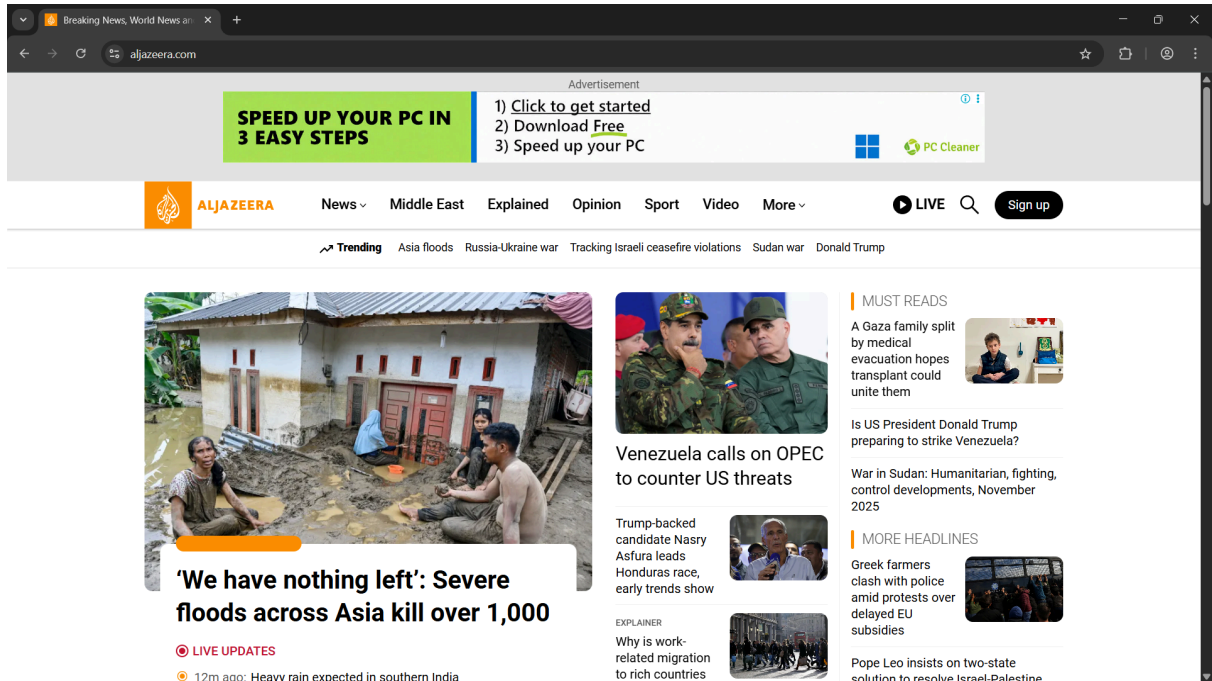
Metric	Value
Total HTTP Requests	142
Total Page Size	2.4 MB (transferred)
DOMContentLoaded	1.93s
Load Time	4.53s
Finish Time	8.44s
Total CSS Requests + (size)	3-req (112KB transferred)
Total JS Requests + (size)	41-req (1,631KB transferred)
Total Image Requests + (size)	59-req (332KB transferred)

1.134 Combined Measurements Table & Average

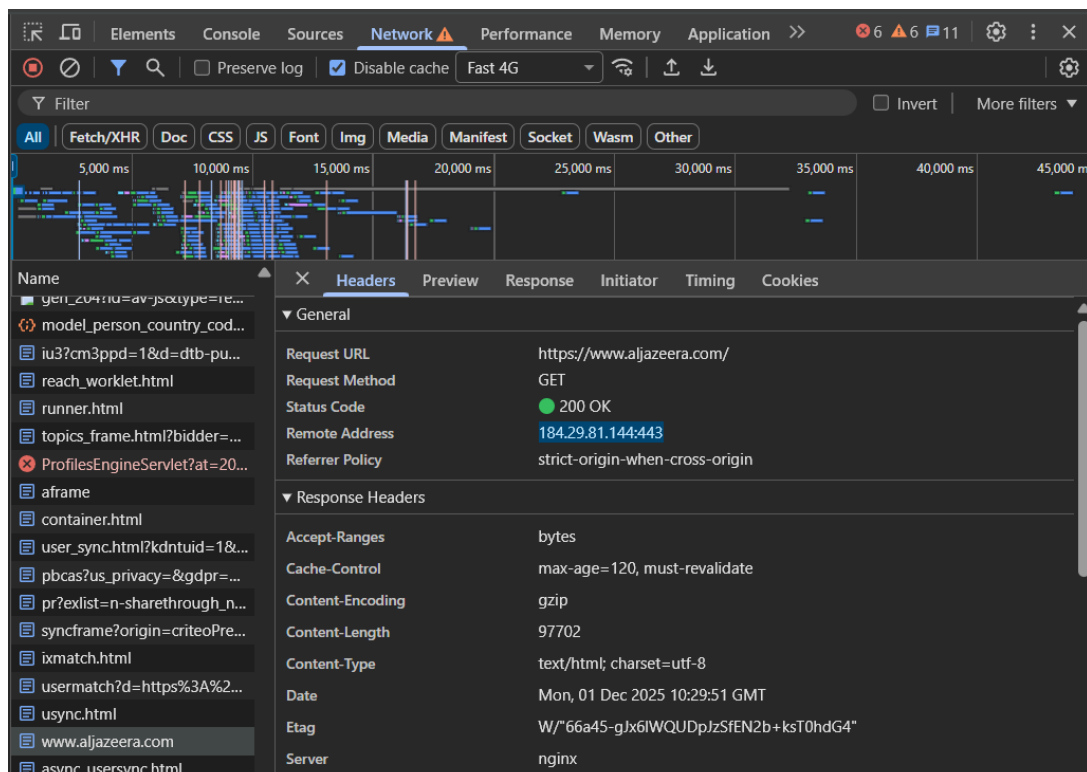
Metric	Measurement-1	Measurement-2	Measurement-3	Average
Total HTTP Requests	145	139	142	142
Total Page Size	2.3 MB (transferred)	2.0 MB (transferred)	2.4 MB (transferred)	2.2MB (transferred)
DOMContentLoaded	1.66s	1.65s	1.93s	1.75s
Load Time	4.21s	3.73s	4.53s	4.15s
Finish Time	13.76s	9.89s	8.44s	10.69s
Total CSS Requests + (size)	3-req (112KB transferred)	3-req (112KB transferred)	3-req (112KB transferred)	3-req (112KB transferred)
Total JS Requests + (size)	41-req (1,631KB transferred)	40-req (1,473KB transferred)	41-req (1,631KB transferred)	45-req (1,578KB transferred)
Total Image Requests + (size)	58-req (229KB transferred)	55-req (134KB transferred)	59-req (332KB transferred)	57-req (232KB transferred)

1.2 Al-Jazeera (News/Media Outlet)

URL - (<https://www.aljazeera.com/>)



1.21 Server Location



Extra - (to find server location)

Geolocation data from **IPRegistry.co** **Product: API, real-time**

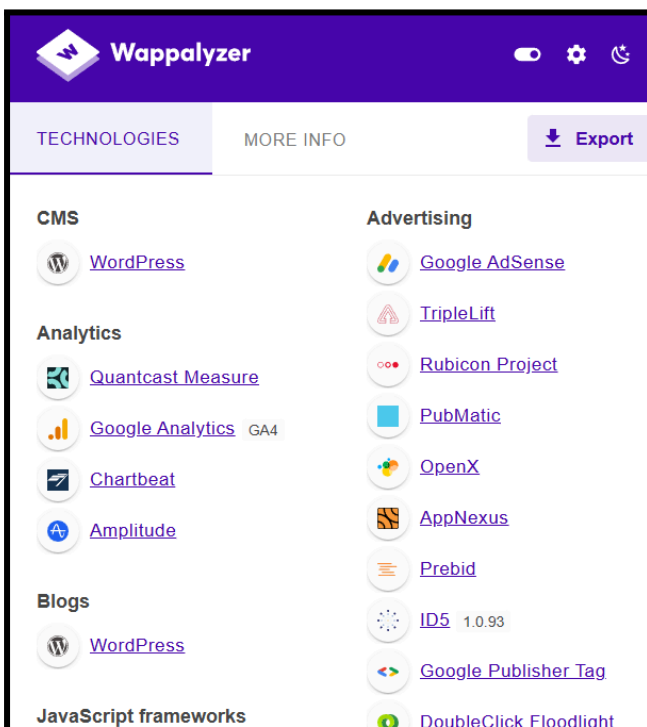
 IP ADDRESS: 184.29.81.144	 ISP: Telecom Italia Sparkle S.P.A.
 COUNTRY: Italy 	 ORGANIZATION: Akamai Technologies, Inc. (akamai.com)
 REGION: Sicilia	 LATITUDE: 38.11569
 CITY: Palermo	 LONGITUDE: 13.36148

Incorrect location? [Contact IPRegistry.co](#)  [view map](#)

Website Name	Organization hosting the Server	Server IP Address	Server Country	Server Region/City
Al-Jazeera	Akamai Technologies	184.29.81.144 (Port 443)	Italy	Sicilia / Palermo

1.22 Technology stack identification (using Both Wappalyzer and DevTools)

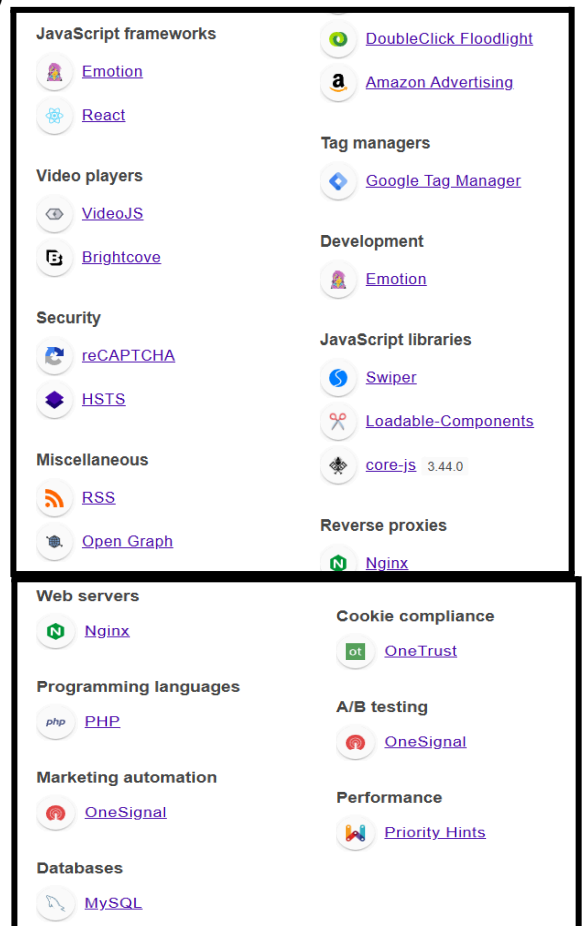
1.221 Wappalyzer:



Wappalyzer

TECHNOLOGIES MORE INFO Export

CMS  WordPress	Advertising  Google AdSense  TripleLift  Rubicon Project  PubMatic  OpenX  AppNexus  Prebid  ID5 1.0.93  Google Publisher Tag
Analytics  Quantcast Measure  Google Analytics GA4  Chartbeat  Amplitude	JavaScript frameworks  DoubleClick Floodlight
Blogs  WordPress	



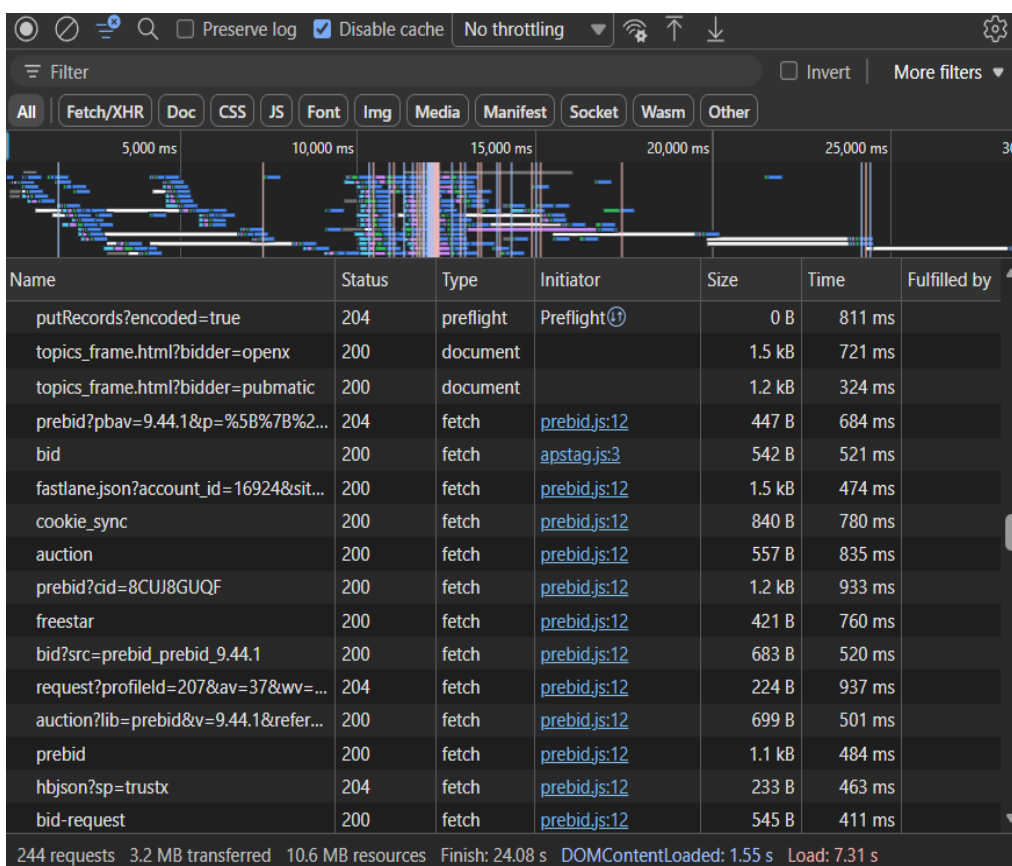
JavaScript frameworks  Emotion  React	 DoubleClick Floodlight  Amazon Advertising
Video players  VideoJS  Brightcove	Tag managers  Google Tag Manager
Security  reCAPTCHA  HSTS	Development  Emotion
Miscellaneous  RSS  Open Graph	JavaScript libraries  Swiper  Loadable-Components  core-js 3.44.0
	Reverse proxies  Nginx
Web servers  Nginx	Cookie compliance  OneTrust
Programming languages  PHP	A/B testing  OneSignal
Marketing automation  OneSignal	Performance  Priority Hints
Databases  MySQL	

Overview:

Content Management System (CMS)	WordPress
Frontend Frameworks	React

1.23 Measurements:

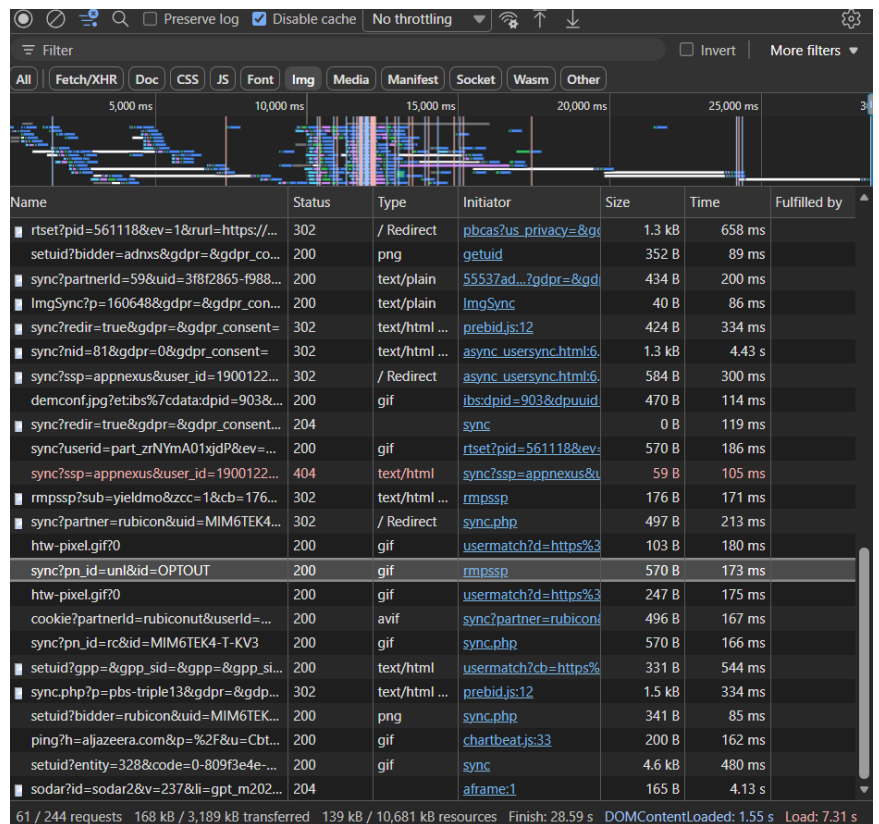
1.231 Measurement 1 (Morning 8 - 9 A.M):



← Page Load Summary

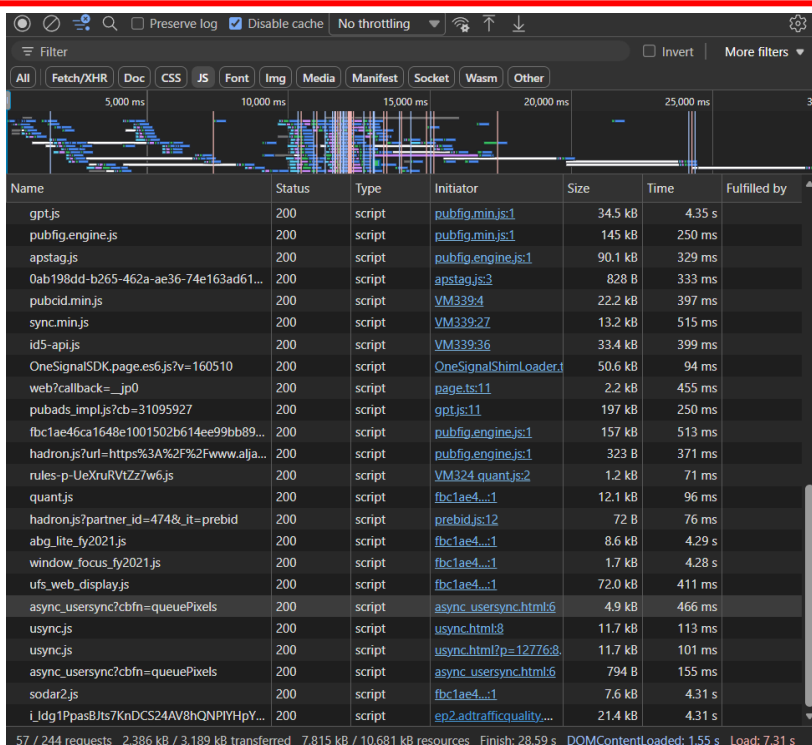
244 requests 3.2 MB transferred 10.6 MB resources Finish: 28.59 s DOMContentLoaded: 1.55 s Load: 7.31 s

Image Load Impact →

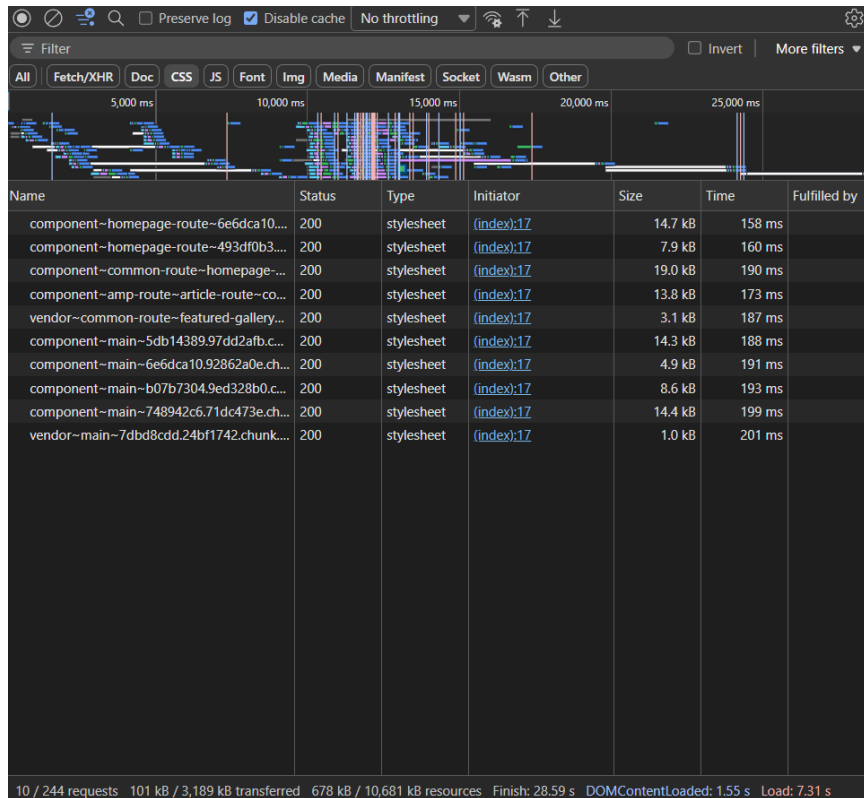


61 / 244 requests 168 kB / 3,189 kB transferred 139 kB / 10,681 kB resources Finish: 28.59 s DOMContentLoaded: 1.55 s Load: 7.31 s

← JS Load Impact



57 / 244 requests 2,386 kB / 3,189 kB transferred 7,815 kB / 10,681 kB resources Finish: 28.59 s DOMContentLoaded: 1.55 s Load: 7.31 s



← CSS Load Impact

10 / 244 requests 101 kB / 3,189 kB transferred 678 kB / 10,681 kB resources Finish: 28.59 s DOMContentLoaded: 1.55 s Load: 7.31 s

1.2311 MSRMNT-1 Table:

Metric	Value
Total HTTP Requests	244
Total Page Size	3.2 MB (transferred)
DOMContentLoaded	1.55s
Load Time	7.31s
Finish Time	28.59s
Total CSS Requests + (size)	10-req (101KB transferred)
Total JS Requests + (size)	57-req (2,386KB transferred)
Total Image Requests + (size)	61-req (168kb transferred)

1.232 Measurement 2 (Afternoon 5 - 6 P.M):

Page Load Summary



237 requests 4.8 MB transferred 12.2 MB resources Finish: 32.16 s DOMContentLoaded: 1.67 s Load: 31.54 s

Image Load Impact



69 / 237 requests 1,930 kB / 4,860 kB transferred 1,906 kB / 12,255 kB resources Finish: 36.13 s DOMContentLoaded: 1.67 s Load: 31.54 s

JS Load Impact



54 / 237 requests 2,332 kB / 4,860 kB transferred 7,694 kB / 12,255 kB resources Finish: 36.13 s DOMContentLoaded: 1.67 s Load: 31.54 s

CSS Load Impact



10 / 237 requests 101 kB / 4,860 kB transferred 678 kB / 12,255 kB resources Finish: 36.13 s DOMContentLoaded: 1.67 s Load: 31.54 s

1.2321 MSRMNT-2 Table:

Metric	Value
Total HTTP Requests	237
Total Page Size	4.8 MB (transferred)
DOMContentLoaded	1.67s
Load Time	31.54s
Finish Time	32.16s
Total CSS Requests + (size)	10-req (101KB transferred)
Total JS Requests + (size)	54-req (2,332KB transferred)
Total Image Requests + (size)	69-req (1,930KB transferred)

1.233 Measurement 3 (Night 11 - 12 P.M):

Page Load Summary



244 requests 5.1 MB transferred 12.8 MB resources Finish: 22.96 s DOMContentLoaded: 2.61 s Load: 10.10 s

Image Load Impact



73 / 244 requests 1,943 kB / 5,064 kB transferred 1,916 kB / 12,800 kB resources Finish: 22.96 s DOMContentLoaded: 2.61 s Load: 10.10 s

JS Load Impact



56 / 244 requests 2,520 kB / 5,064 kB transferred 8,264 kB / 12,800 kB resources Finish: 22.96 s DOMContentLoaded: 2.61 s Load: 10.10 s

CSS Load Impact



10 / 244 requests 101 kB / 5,064 kB transferred 678 kB / 12,800 kB resources Finish: 22.96 s DOMContentLoaded: 2.61 s Load: 10.10 s

1.2331 MSRMNT-3 Table:

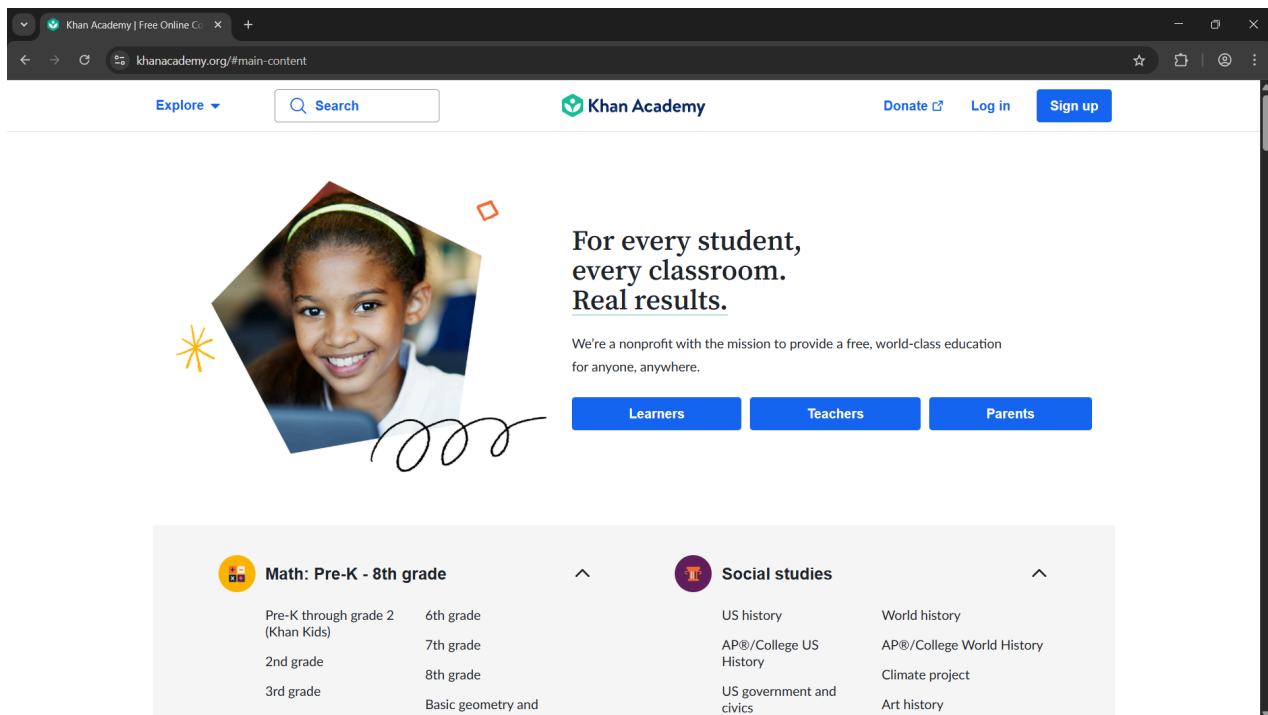
Metric	Value
Total HTTP Requests	244
Total Page Size	5.1 MB (transferred)
DOMContentLoaded	2.61s
Load Time	10.10s
Finish Time	22.96s
Total CSS Requests + (size)	10-req (101KB transferred)
Total JS Requests + (size)	56-req (2,520KB transferred)
Total Image Requests + (size)	73-req (1,943KB transferred)

Note: In Measurement 2 the very high load time that was recorded in the afternoon indicates to us that the server experienced heavier traffic / processing load which has affected performance.

1.234 Combined Measurements Table & Average

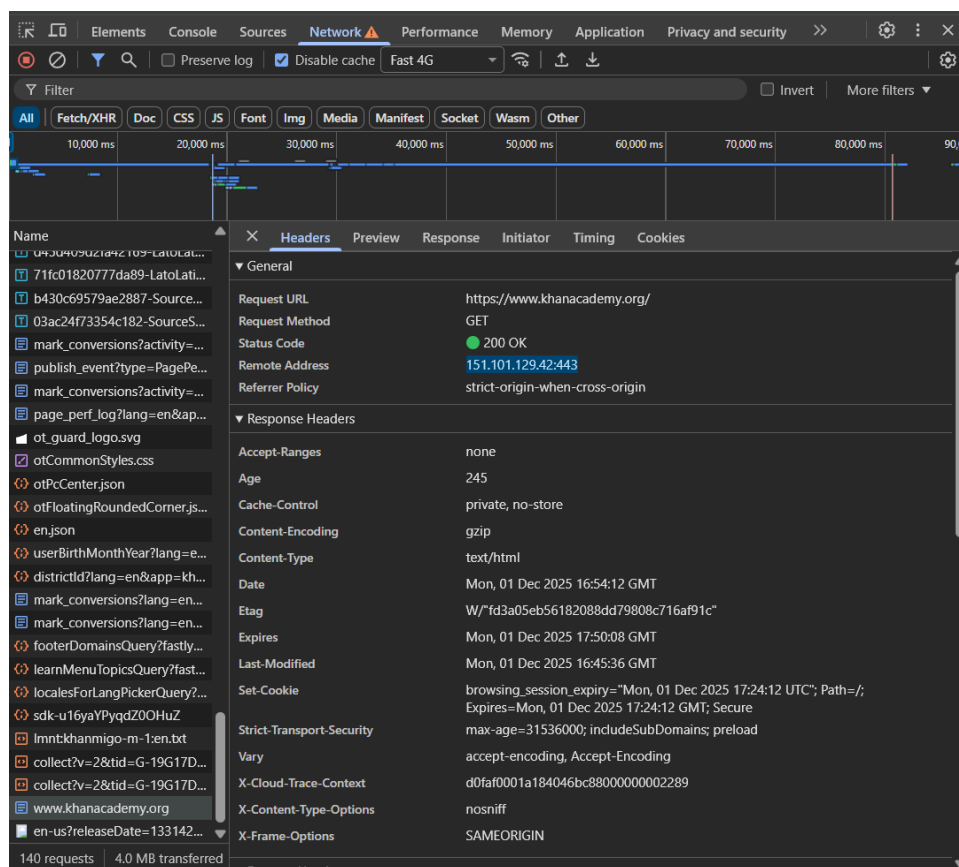
Metric	Measurement-1	Measurement-2	Measurement-3	Average
Total HTTP Requests	244	237	244	241
Total Page Size	3.2 MB (transferred)	4.8 MB (transferred)	5.1 MB (transferred)	4.4 MB (transferred)
DOMContentLoaded	1.55s	1.67s	2.61s	1.94s
Load Time	7.31s	31.54s	10.10s	16.32s
Finish Time	28.59s	32.16s	22.96s	27.9s
Total CSS Requests + (size)	10-req (101KB transferred)	10-req (101KB transferred)	10-req (101KB transferred)	10-req (101KB transferred)
Total JS Requests + (size)	57-req (2,386KB transferred)	54-req (2,332KB transferred)	56-req (2,520KB transferred)	55-req (2,412KB transferred)
Total Image Requests + (size)	61-req (168kb transferred)	69-req (1,930KB transferred)	73-req (1,943KB transferred)	67-req (1347KB transferred)

1.3 Khan Academy (Educational Institution)



URL - (<https://www.khanacademy.org/>)

1.31 Server Location



Extra - (to find server location)

Geolocation data from
IPregistry.co
Product: API, real-time

 **IP ADDRESS:** 151.101.129.42

 **COUNTRY:** United States 

 **REGION:** California

 **CITY:** San Francisco

 **ISP:** Fastly, Inc.

 **ORGANIZATION:** Fastly, Inc. (fastly.com)

 **LATITUDE:** 37.762

 **LONGITUDE:** -122.3972

Incorrect location?
[Contact IPregistry.co](#)
 [view map](#)

Website Name	Organization hosting the Server	Server IP Address	Server Country	Server Region/City
KhanAcademy	Fastly	151.101.129.42 (443)	USA	California / San-Francisco

1.32 Technology stack identification

(using Both Wappalyzer and DevTools)

1.321 Wappalyzer:

 **Wappalyzer**

TECHNOLOGIES
MORE INFO
Export

CMS
 [WordPress](#)

Analytics
 [Google Analytics](#) GA4

Blogs
 [WordPress](#)

JavaScript frameworks
 [React](#)
 [React Router](#) 6

LMS
 [Clever](#)

Programming languages
 [PHP](#)

Databases
 [MySQL](#)

Tag managers
 [Google Tag Manager](#)

Issue trackers
 [Sentry](#)

Security
 [HSTS](#)

Miscellaneous
 [Open Graph](#)

IaaS
 [Google Cloud](#)

Cookie compliance
 [OneTrust](#)

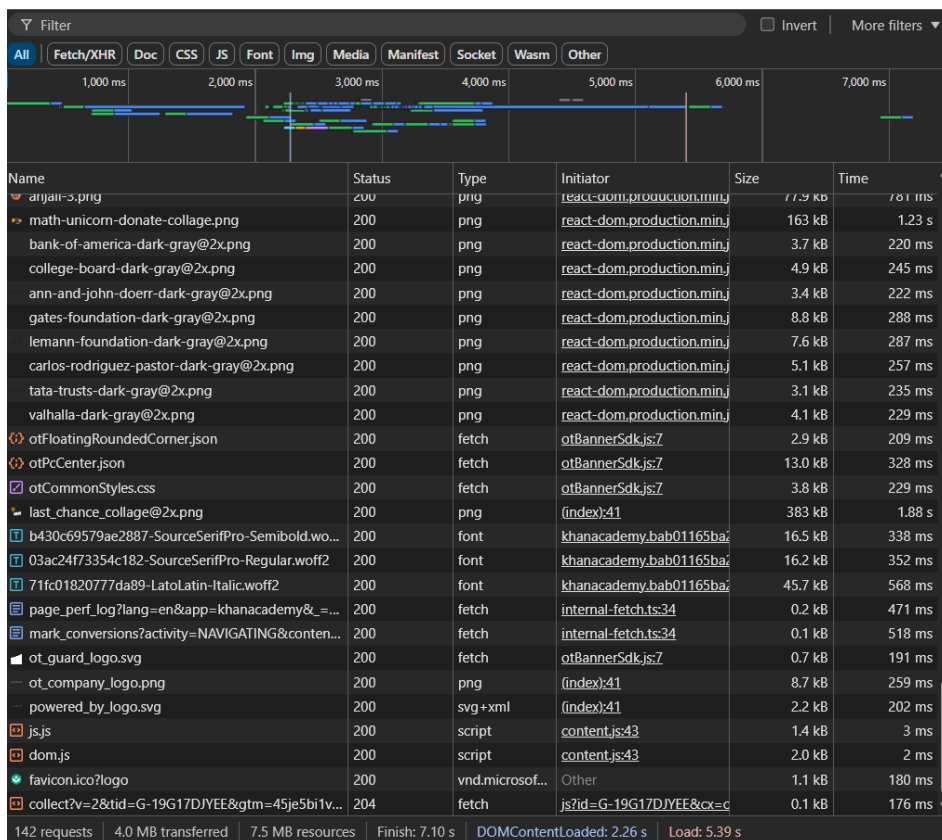
Performance
 [Google Cloud Trace](#)

Overview:

Content Management System (CMS)	WordPress
Frontend Frameworks	React and React Router

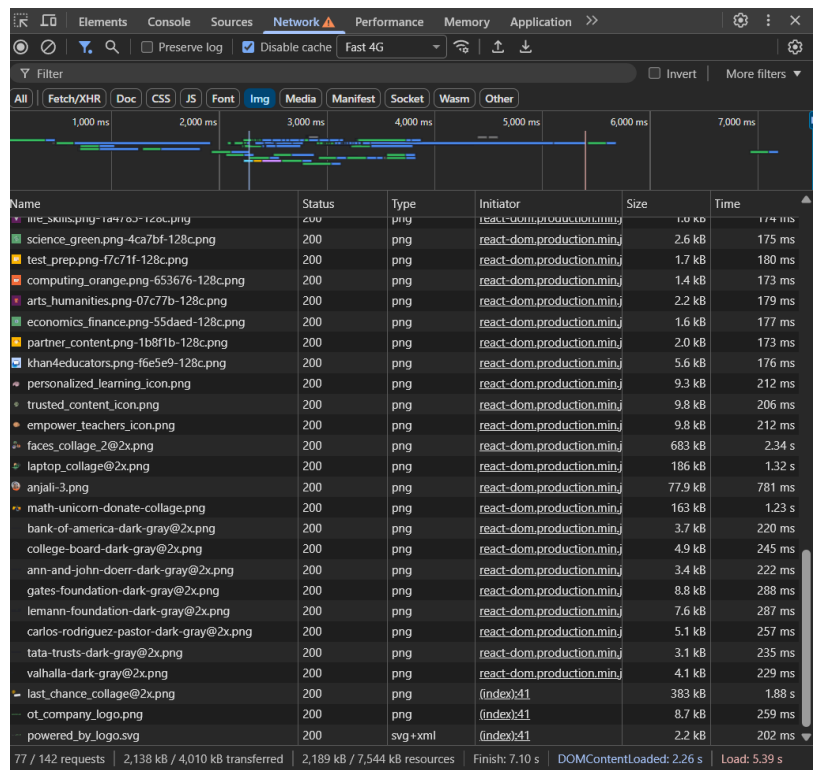
1.33 Measurements:

1.331 Measurement 1 (Morning 8 - 9 A.M):

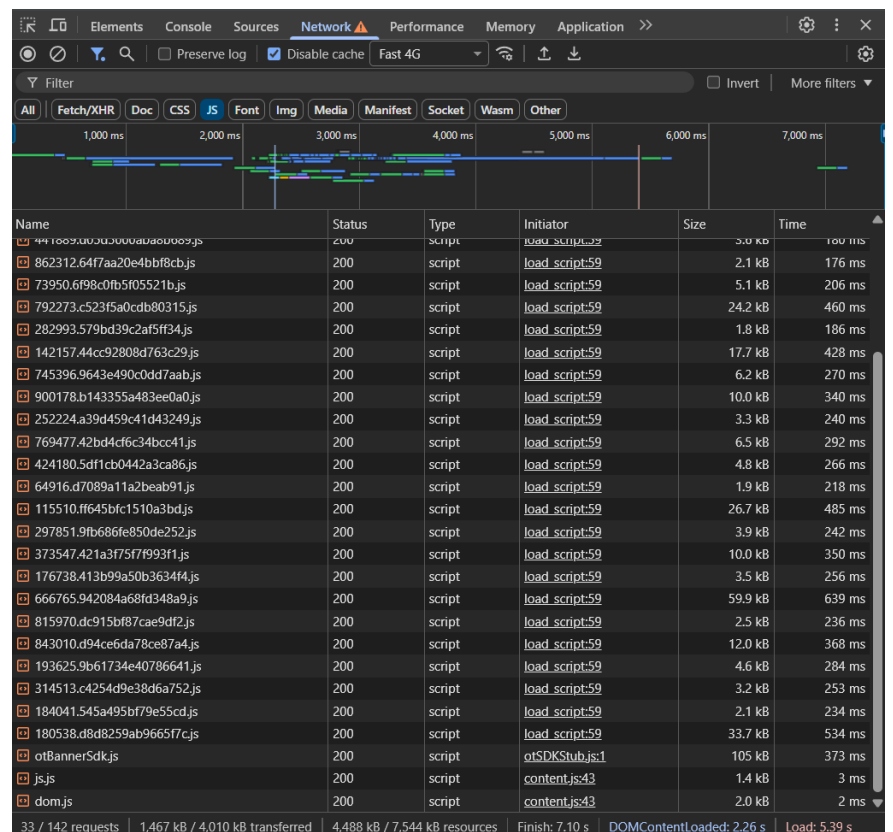


← Page Load Summary

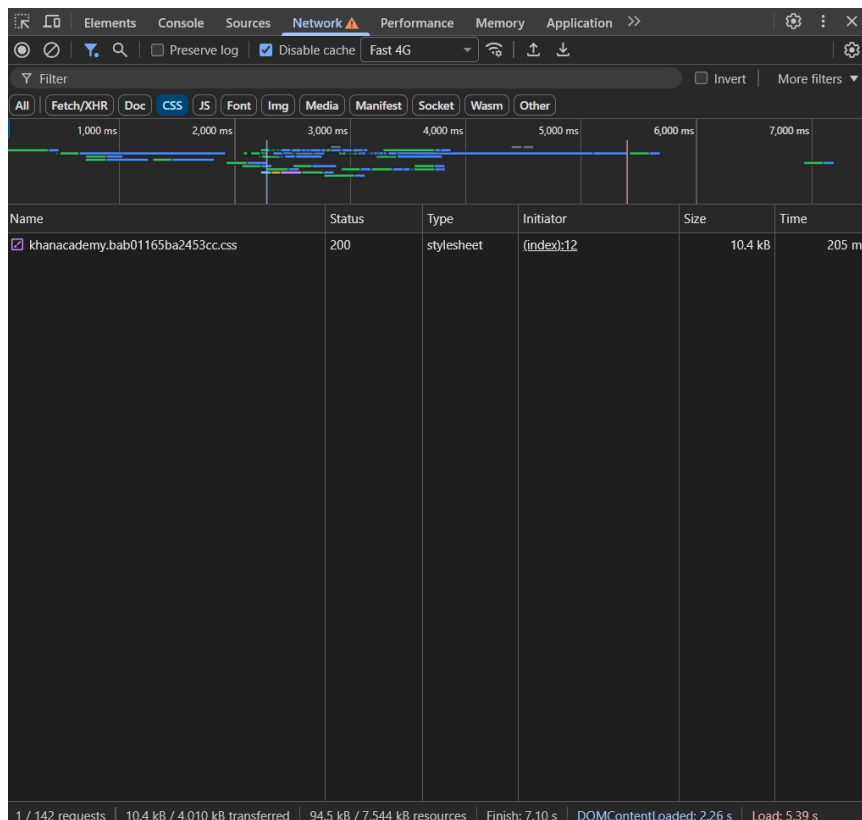
Image Load Impact →



JS Load Impact →



33 / 142 requests | 1,467 kB / 4,010 kB transferred | 4,488 kB / 7,544 kB resources | Finish: 7.10 s | DOMContentLoaded: 2.26 s | Load: 5.39 s



← CSS Load Impact

1 / 142 requests | 10.4 kB / 4,010 kB transferred | 94.5 kB / 7,544 kB resources | Finish: 7.10 s | DOMContentLoaded: 2.26 s | Load: 5.39 s

1.3311 MSRMNT-1 Table:

Metric	Value
Total HTTP Requests	142
Total Page Size	4 MB (transferred)
DOMContentLoaded	2.26s
Load Time	5.39s
Finish Time	7.10s
CSS Files count + (size)	1-req (10.4KB transferred)
JS File count + (size)	33-req (1,467KB transferred)
Image count + (size)	77-req (2,138KB transferred)

1.332 Measurement 2 (Afternoon 5 - 6 P.M):

Page Load Summary



133 requests 4.0 MB transferred 7.6 MB resources Finish: 3.93 s DOMContentLoaded: 1.79 s Load: 3.81 s

Image Load Impact



77 / 133 requests 2,138 kB / 4,024 kB transferred 2,189 kB / 7,613 kB resources Finish: 3.93 s DOMContentLoaded: 1.79 s Load: 3.81 s

JS Load Impact



31 / 133 requests 1,482 kB / 4,024 kB transferred 4,557 kB / 7,613 kB resources Finish: 3.93 s DOMContentLoaded: 1.79 s Load: 3.81 s

CSS Load Impact



1 / 133 requests 10.5 kB / 4,024 kB transferred 94.3 kB / 7,613 kB resources Finish: 3.93 s DOMContentLoaded: 1.79 s Load: 3.81 s

1.3321 MSRMNT-2 Table:

Metric	Value
Total HTTP Requests	133
Total Page Size	4 MB (transferred)
DOMContentLoaded	1.79s
Load Time	3.81s
Finish Time	3.93s
CSS Files count + (size)	1-req (10.5KB transferred)
JS File count + (size)	31-req (1,482KB transferred)
Image count + (size)	77-req (2,138KB transferred)

1.333 Measurement 3 (Night 11 - 12 P.M):

Page Load Summary



134 requests 4.0 MB transferred 7.6 MB resources Finish: 7.21 s DOMContentLoaded: 1.56 s Load: 3.64 s

Image Load Impact



77 / 134 requests 2,139 kB / 4,025 kB transferred 2,189 kB / 7,612 kB resources Finish: 7.21 s DOMContentLoaded: 1.56 s Load: 3.64 s

JS Load Impact



31 / 134 requests 1,483 kB / 4,025 kB transferred 4,557 kB / 7,612 kB resources Finish: 7.21 s DOMContentLoaded: 1.56 s Load: 3.64 s

CSS Load Impact



1 / 134 requests 10.5 kB / 4,025 kB transferred 94.3 kB / 7,612 kB resources Finish: 7.21 s DOMContentLoaded: 1.56 s Load: 3.64 s

1.3331 MSRMNT-3 Table:

Metric	Value
Total HTTP Requests	134
Total Page Size	4 MB (transferred)
DOMContentLoaded	1.56s
Load Time	3.64s
Finish Time	7.21s
Total CSS Requests + (size)	1-req (10.5KB transferred)
Total JS Requests + (size)	31-req (1,483KB transferred)
Total Image Requests + (size)	77-req (2,139KB transferred)

1.334 Combined Measurements Table & Average

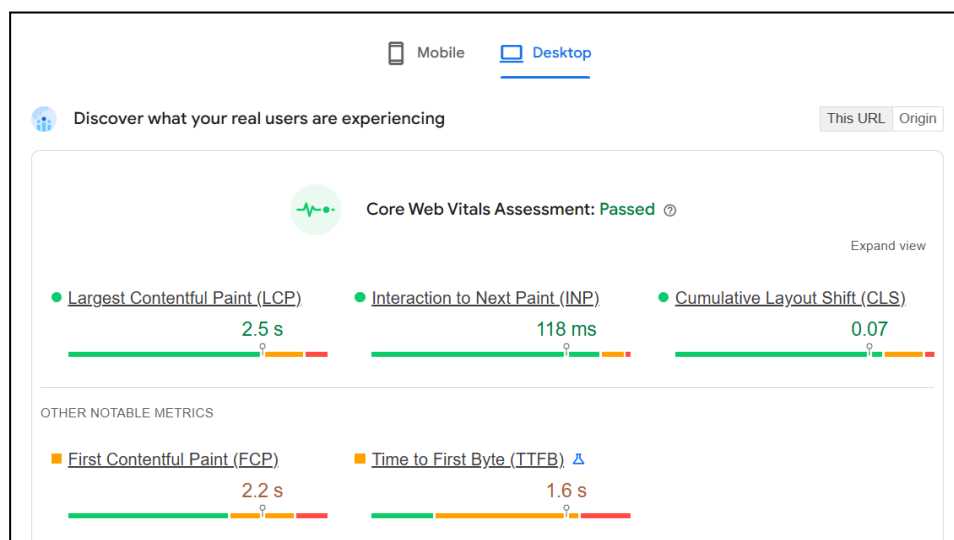
Metric	Measurement-1	Measurement-2	Measurement-3	Average
Total HTTP Requests	142	133	134	136
Total Page Size	4 MB (transferred)	4 MB (transferred)	4 MB (transferred)	4MB (transferred)
DOMContentLoaded	2.26s	1.79s	1.56s	1.87s
Load Time	5.39s	3.81s	3.64s	4.28s
Finish Time	7.10s	3.93s	7.21s	6s
Total CSS Requests + (size)	1-req (10.4KB transferred)	1-req (10.5KB transferred)	1-req (10.5KB transferred)	1-req (10.5KB transferred)
Total JS Requests + (size)	33-req (1,467KB transferred)	31-req (1,482KB transferred)	31-req (1,483KB transferred)	31-req (1477KB transferred)
Total Image Requests + (size)	77-req (2,138KB transferred)	77-req (2,138KB transferred)	77-req (2,139KB transferred)	77-req (2138KB transferred)

2. Core Web Vitals Measurement & Analysis

2.1 Website-1 (Nike)

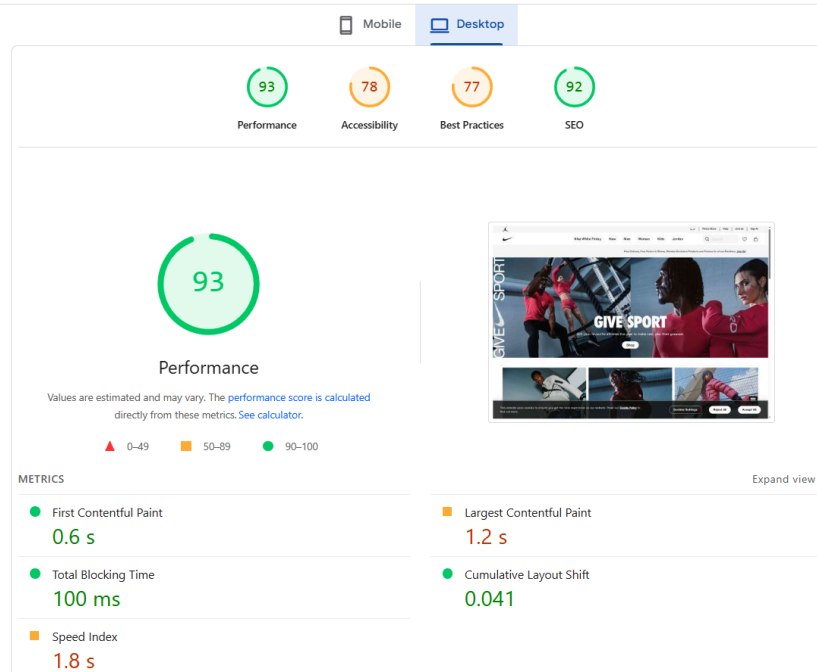
2.1.1 Desktop

Global Data:



Metric	Desktop
Largest Contentful Paint (LCP)	2.5s
Interaction to Next Paint (INP)	118ms
Cumulative Layout Shift (CLS)	0.07

Local (Lab) Data:



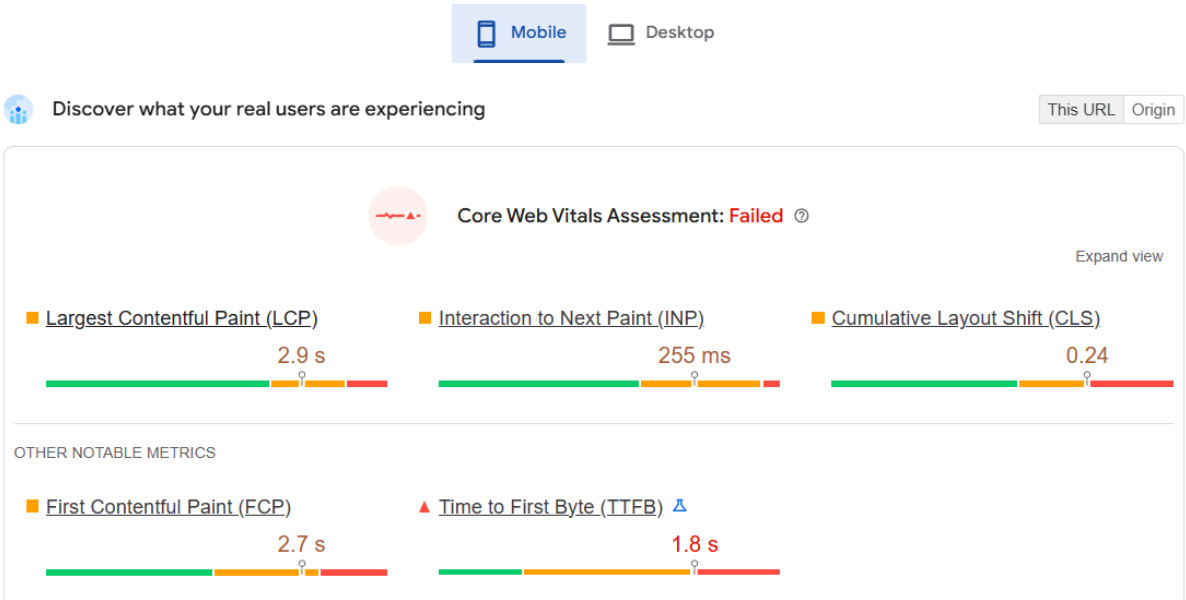
Metric	Desktop
Largest Contentful Paint (LCP)	1.2s
Cumulative Layout Shift (CLS)	0.041

What are Global Measurements and Local (Lab) Measurements?

- Global is where many samples are taken from various network connections using a duration of 28-day period. (Real Visitors over the past month).
- Local is where I measure locally on my Desktop/Mobile (fixed environment). One-Time Measurement using Lighthouse.

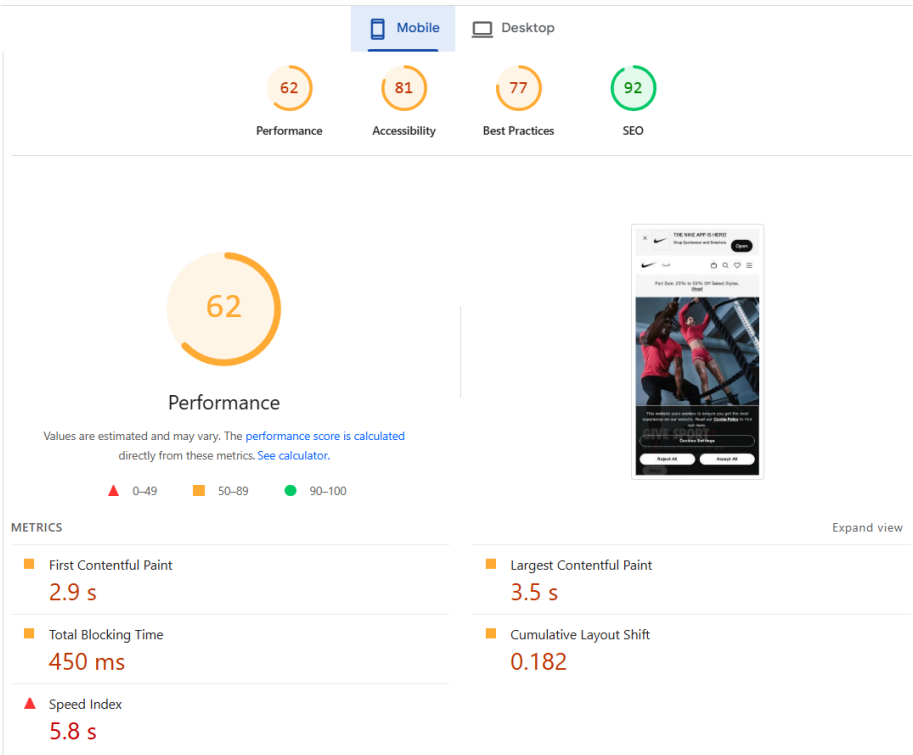
2.1.2 Mobile

Global Data:



Metric	Mobile
Largest Contentful Paint (LCP)	2.9s
Interaction to Next Paint (INP)	255ms
Cumulative Layout Shift (CLS)	0.24

Local
(Lab) Data:



Metric	Mobile
Largest Contentful Paint (LCP)	3.5s
Cumulative Layout Shift (CLS)	0.182

2.1.3 Local Measurements:

- Largest Contentful Paint - (LCP):

Desktop:

The LCP result is 1.2s. This indicates that the website loads the largest content element at 1.2s, which is in the 'Good' time range. (optimal time - Good < 2.5s)

The Specific Element that causes LCP -

```

```

○ LCP breakdown	
Each subpart has specific improvement strategies. Ideally, most of the LCP time should be spent on loading the resources, not within delays. LCP	
Unscored	
Subpart	Duration
Time to first byte	0 ms
Resource load delay	600 ms
Resource load duration	440 ms
Element render delay	120 ms
	Shop Nike's New Gifting Collection 2025 <pre></pre>

Mobile:

The LCP result is 3.5s. This indicates that the website loads the largest content element at 3.5s, which is in the 'Needs Improvement' time range.
(optimal time - Good < 2.5s)

The Specific Element that causes LCP (The same as Desktop) -

```

```

LCP breakdown

Each [subpart has specific improvement strategies](#). Ideally, most of the LCP time should be spent on loading the resources, not within delays. [LCP](#)

Unscored

Subpart	Duration
Time to first byte	0 ms
Resource load delay	780 ms
Resource load duration	50 ms
Element render delay	160 ms



Shop Nike's New Gifting Collection 2025

```

```

Differences and Reasons:

Desktop LCP	Mobile LCP
1.2s - Good	3.5s - Needs Improvement

The differences in LCP are due to mobile devices having much weaker CPU performance overall compared to desktops. For example, when the receiving mobile device gets the JS script files, it will take a much longer time to

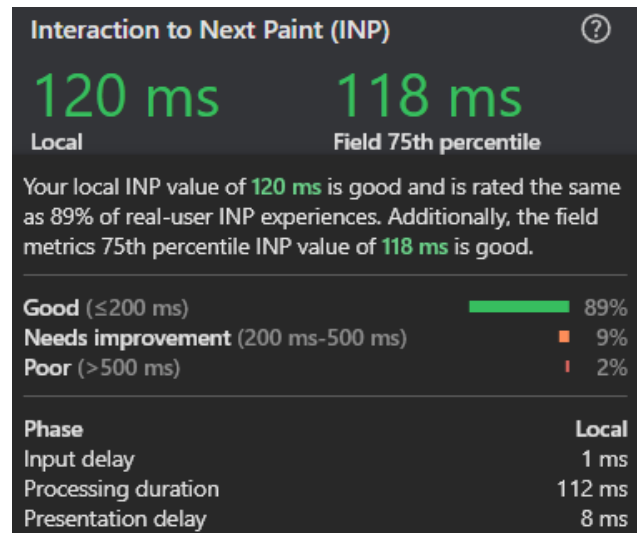
download, parse, compile, and execute them than it would on a desktop.
Lower network speed and throttling can also cause the image to load more slowly on mobile devices.

- Interaction to Next Point - (INP):

Desktop:

Metric	Desktop
Interaction to Next Paint (INP)	120ms

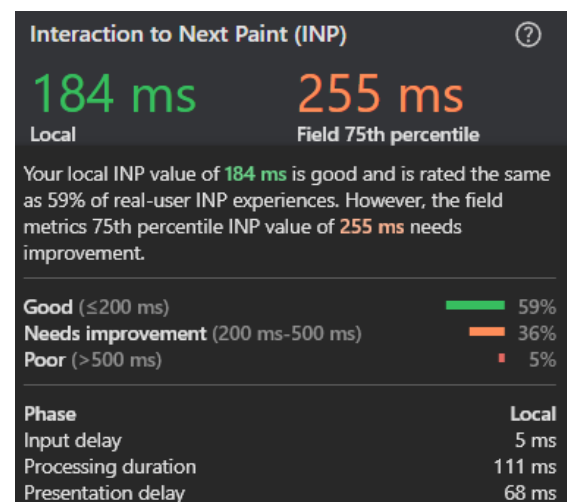
When testing, everything interacted with us instantly. Devtools results show how long each phase took. Input Delay was 1ms, Processing Duration took 112ms, and finally, presentation delay was 8ms. Totaling to 120ms. Which is in the 'Good' time range. (optimal time - Good < 200ms)



Mobile:

Metric	Mobile
Interaction to Next Paint (INP)	184ms

When testing, everything interacted with us finely without any noticeable delay. Devtools results show how long each phase took. Input Delay was 5ms, Processing Duration took 111ms, and finally, presentation delay was 68ms. Totaling to 184ms. Which is in the 'Good' time range. (optimal time - Good < 200ms). But as shown in the image that many real-world users (75th percentile of them) have experienced 255ms (Global Data), suggesting that a slight improvement is needed so that, in the worst possible case, interaction speed doesn't go over 200ms.



- Cumulative Layout Shift - (CLS):

Desktop:

Metric	Desktop
Cumulative Layout Shift (CLS)	0.041

The CLS result obtained is 0.041, which indicates very little shifting when loading. (Good < 0.1)

Some images are “Unsize image element”, which affects the CLS. There is also a Cookie Banner, which is dynamically added after page load, increasing the CLS.

INSIGHTS

Mobile Desktop

Layout shift culprits

Layout shifts occur when elements move absent any user interaction. Investigate the causes of layout shifts, such as elements being added, removed, or their fonts changing as the page loads. [CLS] [Unscored]

Element	Layout shift score
Total	0.041
 Nike UAE Official Website - Shop Shoes, Clothing & Accessories GIVE SPORT G... <main class="b-page__main" id="maincontent">	0.040
 Shop Nike's New Gifting Collection 2025 Unsize image	Unsize image
 Nike White Friday New Men Women Kids Jordan <div class="b-megamenu__holder">	0.000
 Shop Nike's New Gifting Collection 2025 Unsize image	Unsize image
 This website uses cookies to ensure you get the best experience on our website... <div id="onetrust-banner-sdk" class="otFlat bottom ot-wo-title vertical-align-content focus-visible" tabindex="0" role="region" aria-label="Cookie banner" style="bottom: 0px;" data-focus-visible-added="">	0.000

The Elements that cause CLS -

```
<main class="b-page__main" id="maincontent">
```

```
 - (Hero Banner)
```

Mobile:

Metric	Mobile
Cumulative Layout Shift (CLS)	0.182

The CLS result obtained is 0.182, which indicates there is shifting but it is not severe. It is in the 'Needs Improvement' range. (Good < 0.1)

Element	Layout shift score
Total	0.182
 Nike UAE Official Website - Shop Shoes, Clothing & Accessories GIVE SPORT Gif... <main class="b-page__main" id="maincontent">	0.100
 Shop Nike's New Gifting Collection 2025 	0.074
 Nike UAE Official Website - Shop Shoes, Clothing & Accessories GIVE SPORT Gif... <main class="b-page__main" id="maincontent">	0.001
 Shop Nike's New Gifting Collection 2025 	
 This website uses cookies to ensure you get the best experience on our website... <div id="onetrust-banner-sdk" class="otFlat bottom ot-wo-title ot-buttons-fw focus-visible" tabindex="0" role="region" aria-label="Cookie banner" style="bottom: 0px;" data-focus-visible-added="">	

The Main Banner image Layout Shift Score is 0.100, and then at a later loading stage, it got a Layout Shift Score of 0.074 - (Hero Banner). By itself, it got a Layout Shift Score of 0.174, which is due to it loading in parts, which is why it has multiple scores.

The Elements that cause CLS -

1. <main class="b-page__main" id="maincontent">
(Layout Shift Score 0.100)
2.
(Layout Shift Score 0.074)
3. <div id="onetrust-banner-sdk" class="otFlat bottom ot-wo-title ot-buttons-fw focus-visible" tabindex="0" role="region"

```
aria-label="Cookie banner" style="bottom: 0px;"  
data-focus-visible-added="">  
(Layout Shift Score 0.001)
```

- Mobile vs Desktop Performance Comparison Table

Metric	Mobile	Desktop
Performance Score	62	93
Largest Contentful Paint (LCP)	3.5s (Needs Improvement)	1.2s (Good)
Interaction to Next Paint (INP)	184ms (Good)	120ms (Good)
Cumulative Layout Shift (CLS)	0.182 (Needs Improvement)	0.041 (Good)

Overall, the desktop version of the website is significantly better in all metrics, reaching 'Good Ranges' where no improvements are needed. The mobile version has 2 areas that need to be improved (LCP, CLS), which indicates that there is slower loading for the largest visible content and layout instability.

Top 3 Issues Flagged by Lighthouse

Desktop:

Issue	Type	Savings
Reduced Unused JS	JS	627 kib
Improve image Delivery	Images	371 kib
Use efficient cache lifetimes	Cache	346 kib

INSIGHTS
▲ Use efficient cache lifetimes — Est savings of 346 KiB
▲ Improve image delivery — Est savings of 371 KiB
▲ Render blocking requests — Est savings of 410 ms
▲ Font display — Est savings of 340 ms
▲ LCP request discovery
▲ Forced reflow
▲ Network dependency tree
■ Legacy JavaScript — Est savings of 41 KiB
○ Layout shift culprits
○ LCP breakdown
○ 3rd parties

DIAGNOSTICS
▲ Reduce unused CSS — Est savings of 115 KiB
▲ Reduce unused JavaScript — Est savings of 627 KiB
■ Image elements do not have explicit <code>width</code> and <code>height</code>

Mobile:

Issue	Type	Savings
Reduced Unused JS	JS	631 kib
Render-Blocking Requests	CSS/JS	2190 ms
Efficient cache lifetimes	Cache	347 kib

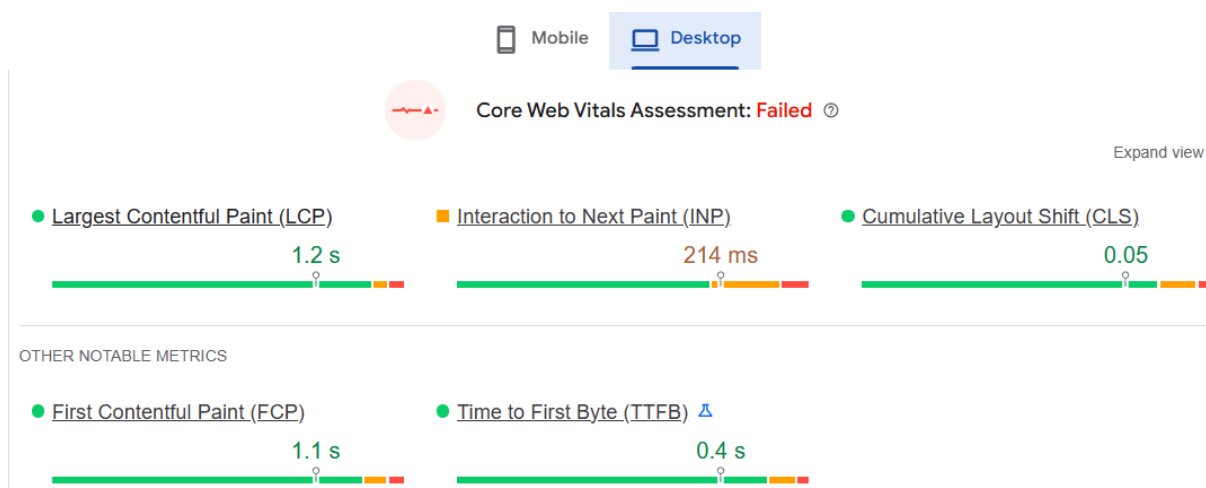
INSIGHTS
▲ Render blocking requests — Est savings of 2,190 ms
▲ Font display — Est savings of 350 ms
▲ Use efficient cache lifetimes — Est savings of 347 KiB
▲ Layout shift culprits
▲ LCP request discovery
▲ Network dependency tree
■ Legacy JavaScript — Est savings of 41 KiB
○ LCP breakdown
○ 3rd parties

DIAGNOSTICS
▲ Reduce JavaScript execution time — 1.7 s
▲ Minimize main-thread work — 2.7 s
▲ Reduce unused CSS — Est savings of 118 KiB
■ Image elements do not have explicit <code>width</code> and <code>height</code>
■ Reduce unused JavaScript — Est savings of 631 KiB
○ Avoid long main-thread tasks — 10 long tasks found
○ Avoid non-composited animations — 1 animated element found
○ User Timing marks and measures — 2 user timings

2.2 Website-2 (Al-Jazeera)

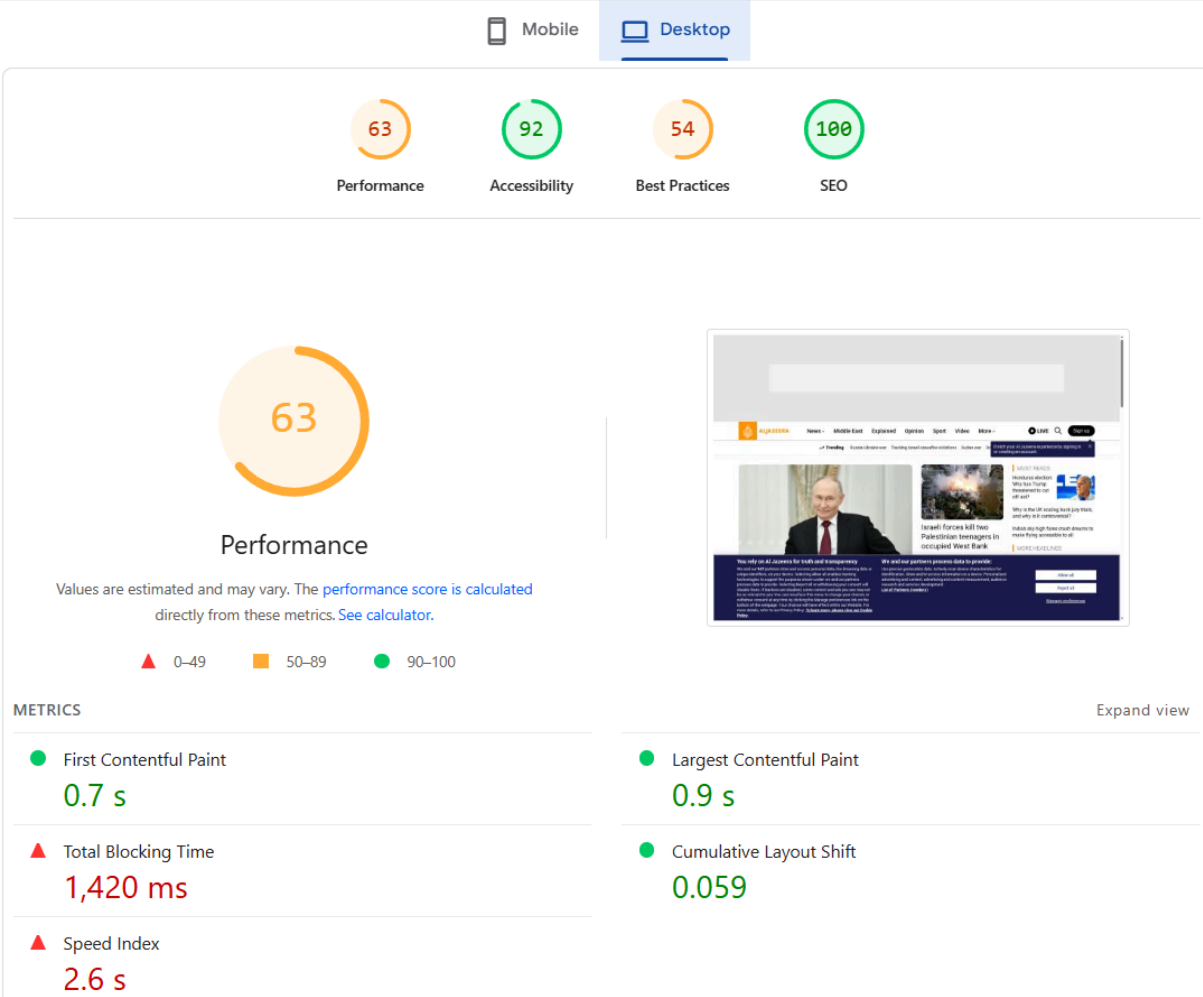
2.2.1 Desktop

Global Data:



Metric	Desktop
Largest Contentful Paint (LCP)	1.2s
Interaction to Next Paint (INP)	214ms
Cumulative Layout Shift (CLS)	0.05

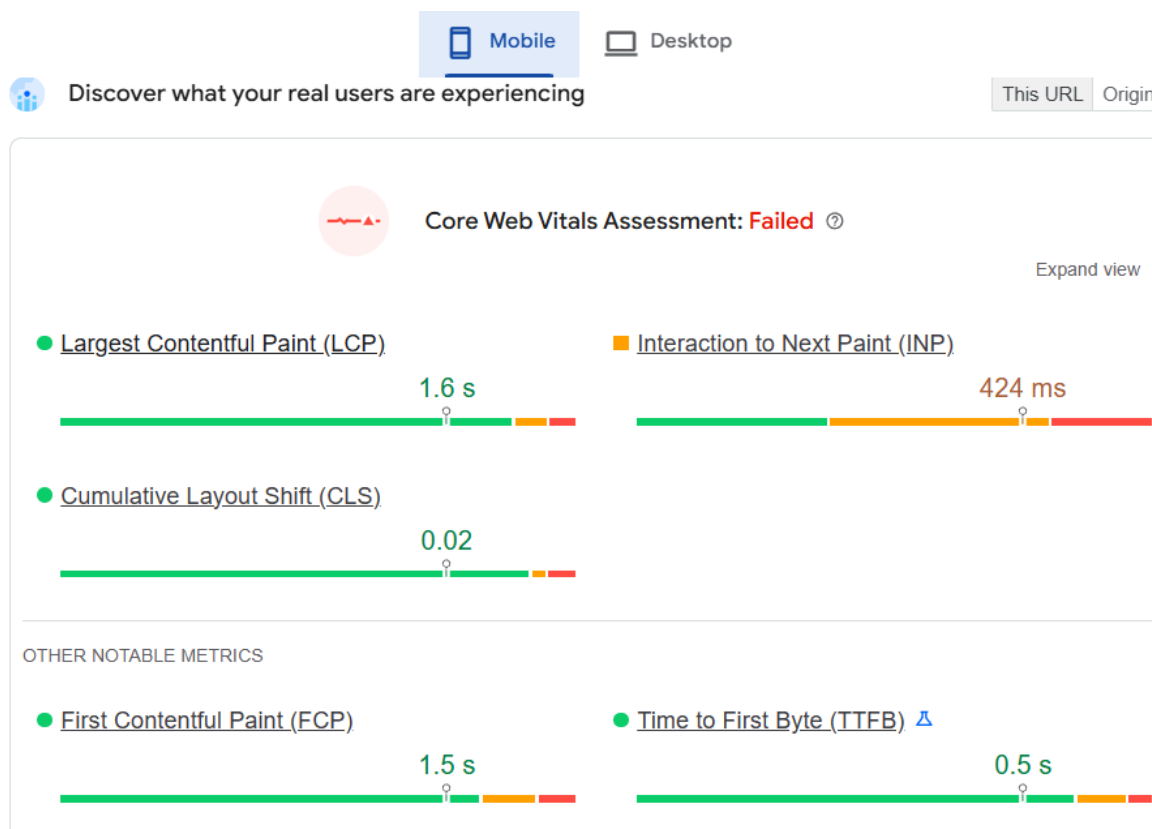
Local (Lab) Data:



Metric	Desktop
Largest Contentful Paint (LCP)	0.9s
Cumulative Layout Shift (CLS)	0.059

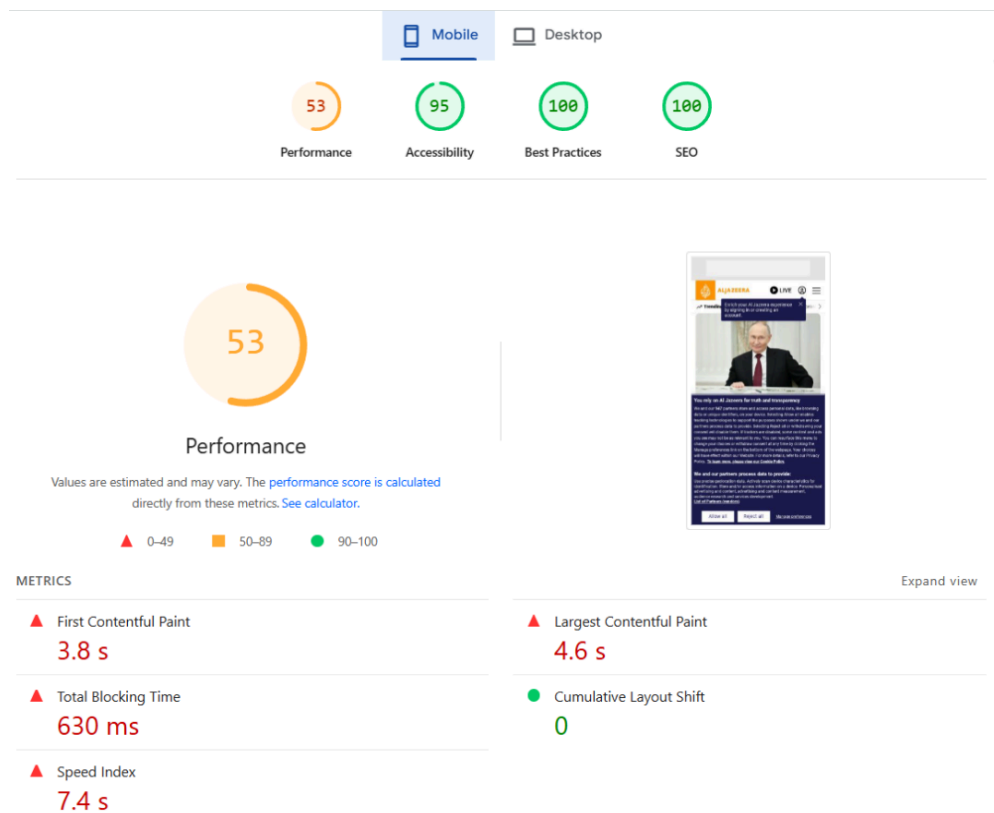
2.2.2 Mobile

Global Data:



Metric	Mobile
Largest Contentful Paint (LCP)	1.6s
Interaction to Next Paint (INP)	424ms
Cumulative Layout Shift (CLS)	0.02

Local
(Lab) Data:



Metric	Mobile
Largest Contentful Paint (LCP)	4.6s
Cumulative Layout Shift (CLS)	0

2.2.3 Local Measurements:

- Largest Contentful Paint - (LCP):

Desktop:

The LCP result is 0.9s. This indicates that the website loads the largest content element at 0.9s, which is in the 'Good' time range. (optimal time - Good < 2.5s).

The Specific Element that causes LCP -

```

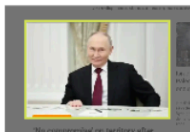
```

LCP breakdown

Each subpart has specific improvement strategies. Ideally, most of the LCP time should be spent on loading the resources, not within delays. [LCP](#)

(Unscored)

Subpart	Duration
Time to first byte	0 ms
Resource load delay	80 ms
Resource load duration	40 ms
Element render delay	160 ms



agency Sputnik, R...>

In this pool photograph distributed by the Russian state agency Sputnik, Russia...

```

```

Mobile:

The LCP result is 4.6s. This indicates that the website loads the largest content element at 4.6s, which is in the 'Poor' time range. (optimal time needed - Good < 2.5s)

The Specific Element that causes LCP (The same as Desktop) -

```

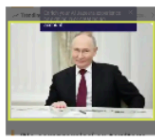
```

○ LCP breakdown

Each [subpart has specific improvement strategies](#). Ideally, most of the LCP time should be spent on loading the resources, not within delays. [LCP](#)

Unscored

Subpart	Duration
Time to first byte	0 ms
Resource load delay	100 ms
Resource load duration	10 ms
Element render delay	170 ms



Sputnik, R...>

In this pool photograph distributed by the Russian state agency Sputnik, Russia...

```
<img class="articles-card_image" loading="eager" fetchpriority="high" src="/wp-content/uploads/2025/12/afp_692f72a4042e-1764717220.jpg?resize=570%2C3..." srcset="/wp-content/uploads/2025/12/afp_692f72a4042e-1764717220.jpg?resize=140%2C9..." sizes="(max-width: 140px) 140px, (max-width: 170px) 170px, (max-width: 270px) 270..." alt="In this pool photograph distributed by the Russian state agency
```

Differences and Reasons:

Desktop LCP	Mobile LCP
0.9s - Good	4.6s - Poor

Desktop LCP (0.9s) is a lot faster than Mobile LCP (4.6s).

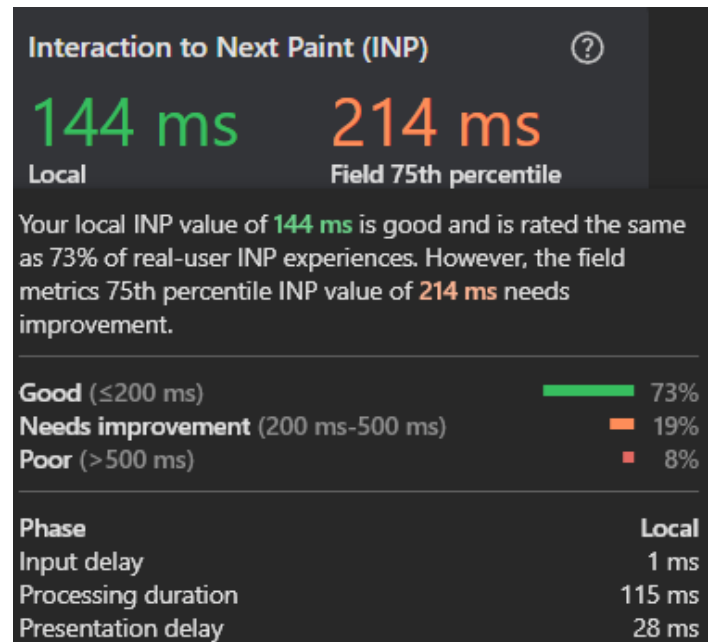
The reasons for that to happen are many. Device Performance, Large Image Size (3,452 kib), Network Throttling, and JS execution may be the cause of LCP. Using the images of LCP breakdown we can see that the element render delay is higher on mobile reaching 170ms while desktop takes 160 ms (note: the task is handled much faster because of a stronger cpu).

Interaction to Next Point - (INP):

Desktop:

Metric	Desktop
Interaction to Next Paint (INP)	144ms

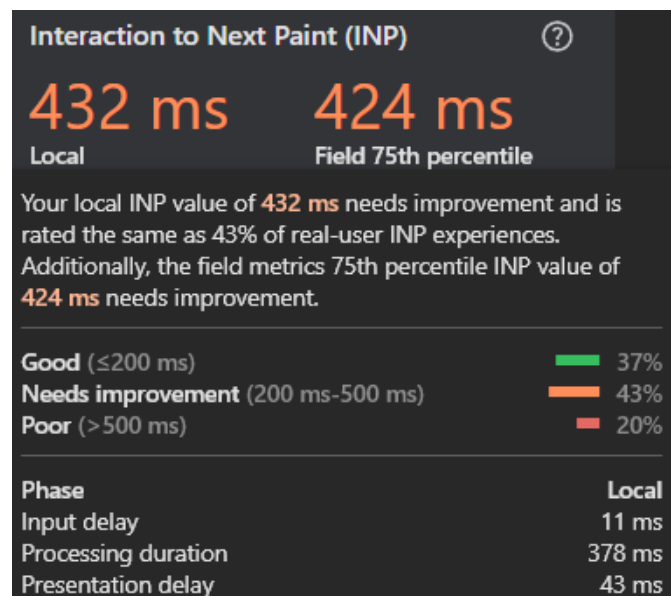
When testing, everything interacted with us instantly. Devtools results show how long each phase took. Input Delay was 1ms, Processing Duration took 115ms, and finally, presentation delay was 28ms. Totaling to 144ms. Which is in the 'Good' time range. (optimal time - Good < 200ms)



Mobile:

Metric	Mobile
Interaction to Next Paint (INP)	432ms

When testing, everything interacted with us finely, but with noticeable delay. Devtools results show how long each phase took. Input Delay was 11ms, Processing Duration took 378ms, and finally, presentation delay was 43ms. Totaling to 432ms. Which is in the 'Needs Improvement' time range. (optimal time - Good < 200ms). Improvements should be made so that the User experience is positive.



- Cumulative Layout Shift - (CLS):

Desktop:

Metric	Desktop
Cumulative Layout Shift (CLS)	0.059

The CLS result obtained is 0.059, which indicates very little shifting when loading. (Good < 0.1)

The only CLS found was from a Cookie consent banner, which is a Dynamic Element

Element	Layout shift score
Total	0.059
 You rely on Al Jazeera for truth and transparency We and our 947 partners store... <div id="onetrust-banner-sdk" class="otFlat ot-iab-2 bottom vertical-align-content ot-buttons-fw" tabindex="0" role="region" aria-label="Cookie banner">	0.056
 You rely on Al Jazeera for truth and transparency We and our 947 partners store... <div id="onetrust-banner-sdk" class="otFlat ot-iab-2 bottom vertical-align-content ot-buttons-fw" tabindex="0" role="region" aria-label="Cookie banner">	0.003

The Element that causes CLS -

```
<div id="onetrust-banner-sdk" class="otFlat ot-iab-2 bottom
vertical-align-content ot-buttons-fw" tabindex="0" role="region"
aria-label="Cookie banner">
```

Mobile:

Metric	Mobile
Cumulative Layout Shift (CLS)	0

The CLS result obtained is 0.0, which indicates no shifting at all when loading 'Good range'. (Good < 0.1)

● Mobile vs Desktop Performance Comparison Table

Metric	Mobile	Desktop
Performance Score	53	63
Largest Contentful Paint (LCP)	4.6s (Poor)	0.9s (Good)
Interaction to Next Paint (INP)	432ms (Needs Improvement)	144ms (Good)
Cumulative Layout Shift (CLS)	0.0 (Good)	0.059 (Good)

Overall, the desktop version of the website is significantly better in all metrics, reaching ‘Good Ranges’ where no improvements are needed. The mobile version has 1 area that needs to be fixed instantly (LCP), which indicates that there is a significantly long loading time for the largest visible content, and 1 area that needs to be improved (INP) for smoother interaction, which increases User Experience.

Top 3 Issues Flagged by Lighthouse

Desktop:

Issue	Type	Savings
Reduced Unused JS	JS	1,234 kib
Improve image Delivery	Images	915 kib
Use efficient cache lifetimes	Cache	699 kib

INSIGHTS	DIAGNOSTICS
▲ Use efficient cache lifetimes — Est savings of 699 KiB ▲ Forced reflow ■ Improve image delivery — Est savings of 915 KiB ■ Legacy JavaScript — Est savings of 42 KiB ○ Layout shift culprits ○ LCP breakdown ○ 3rd parties	▲ Minimize main-thread work — 4.8 s ▲ Reduce JavaScript execution time — 2.5 s ▲ Reduce unused JavaScript — Est savings of 1,234 KiB ▲ Reduce unused CSS — Est savings of 105 KiB ■ Avoid enormous network payloads — Total size was 5,426 KiB ○ Avoid long main-thread tasks — 16 long tasks found ○ User Timing marks and measures — 21 user timings

Mobile:

Issue	Type	Savings
Improve Image Delivery	Images	3,425 kib
Reduce Unused JS	JS	906 kib
Efficient cache lifetimes	Cache	719 kib

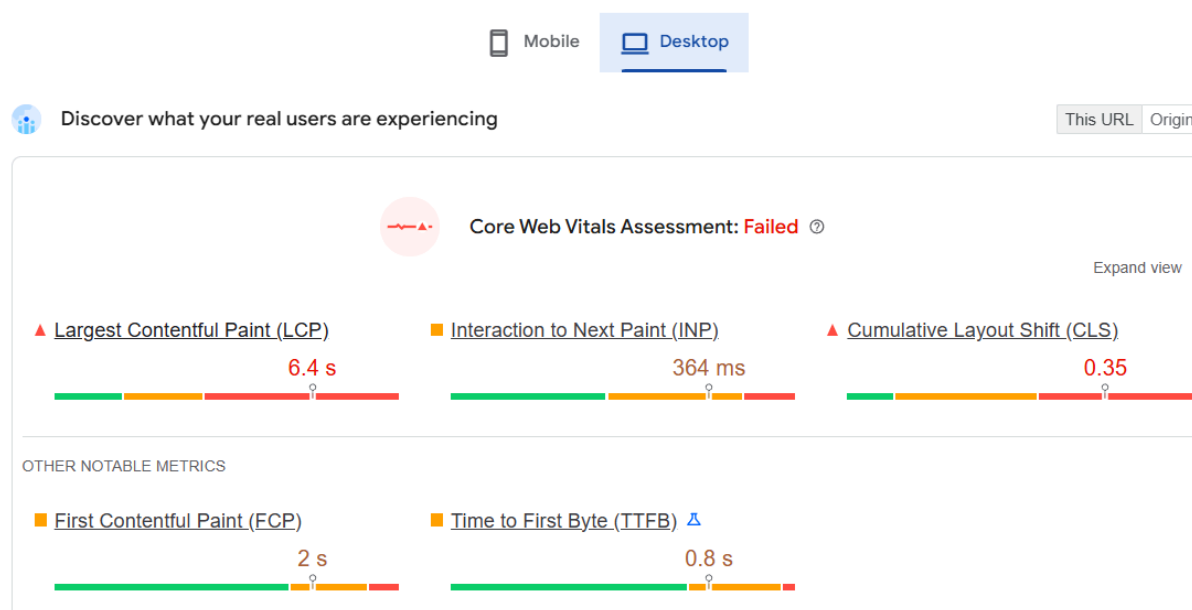
INSIGHTS
▲ Improve image delivery — Est savings of 3,425 KiB
▲ Use efficient cache lifetimes — Est savings of 719 KiB
▲ Legacy JavaScript — Est savings of 42 KiB
▲ Forced reflow
○ Optimize DOM size
○ LCP breakdown
○ 3rd parties

DIAGNOSTICS
▲ Reduce JavaScript execution time — 4.6 s
▲ Minimize main-thread work — 7.6 s
▲ Reduce unused JavaScript — Est savings of 906 KiB
▲ Reduce unused CSS — Est savings of 69 KiB
■ Avoid enormous network payloads — Total size was 6,989 KiB
○ Avoid long main-thread tasks — 20 long tasks found
○ Avoid non-composited animations — 2 animated elements found
○ User Timing marks and measures — 214 user timings

2.3 Website-3 (KhanAcademy)

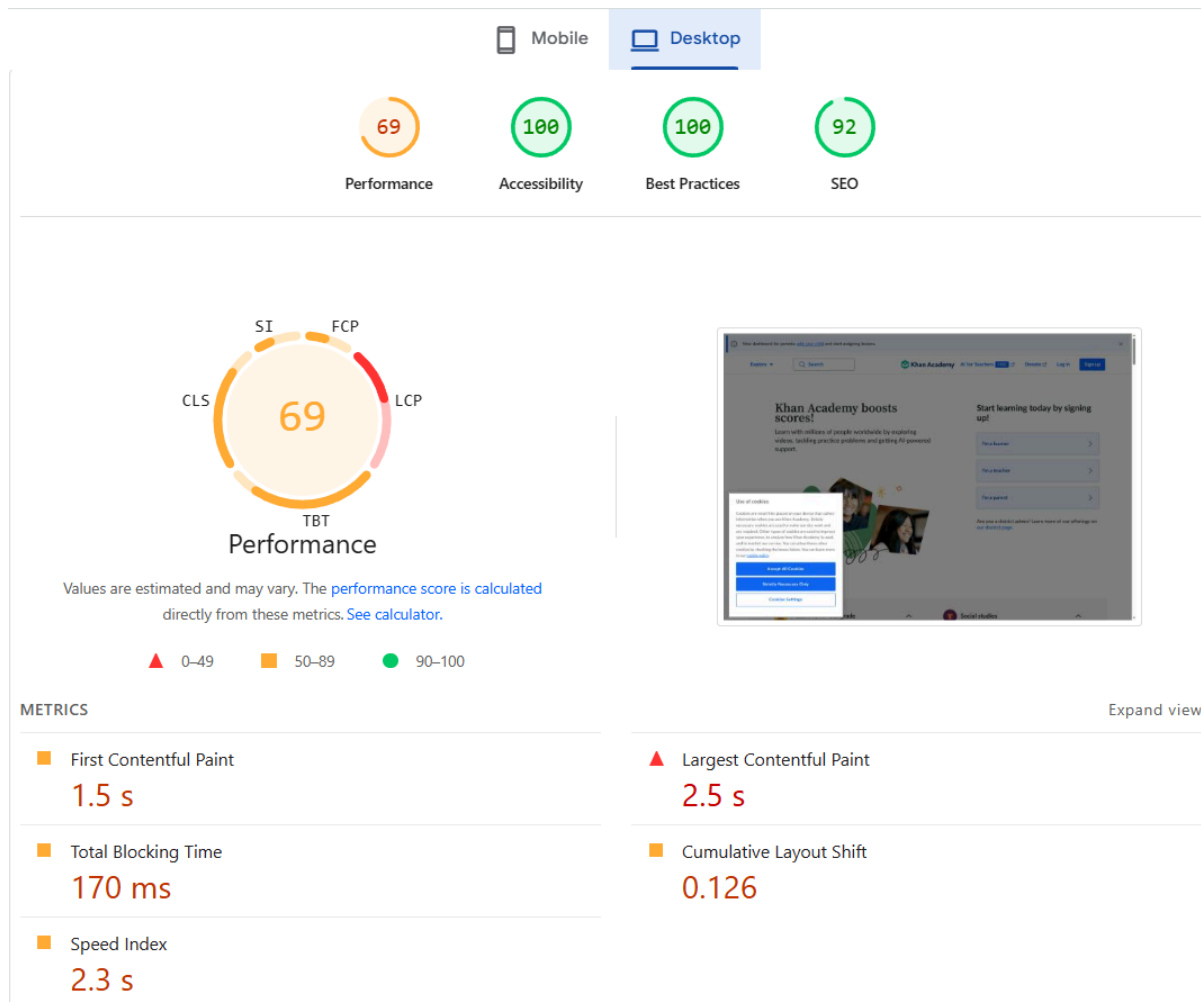
2.3.1 Desktop

Global Data:



Metric	Desktop
Largest Contentful Paint (LCP)	6.4s
Interaction to Next Paint (INP)	364ms
Cumulative Layout Shift (CLS)	0.35

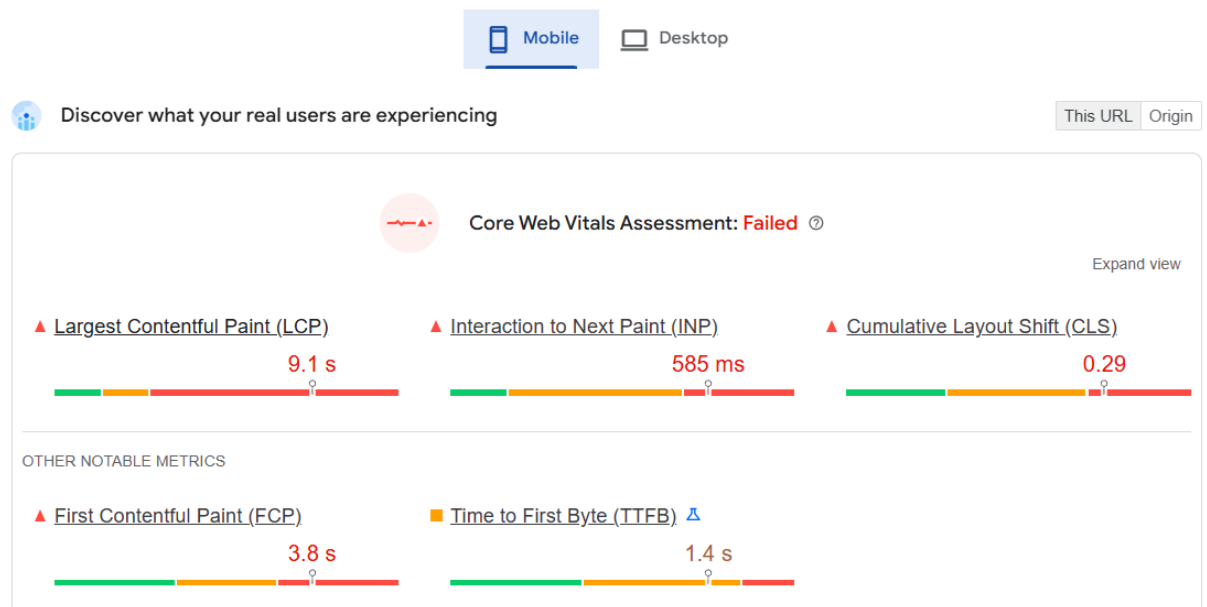
Local (Lab) Data:



Metric	Desktop
Largest Contentful Paint (LCP)	2.5s
Cumulative Layout Shift (CLS)	0.126

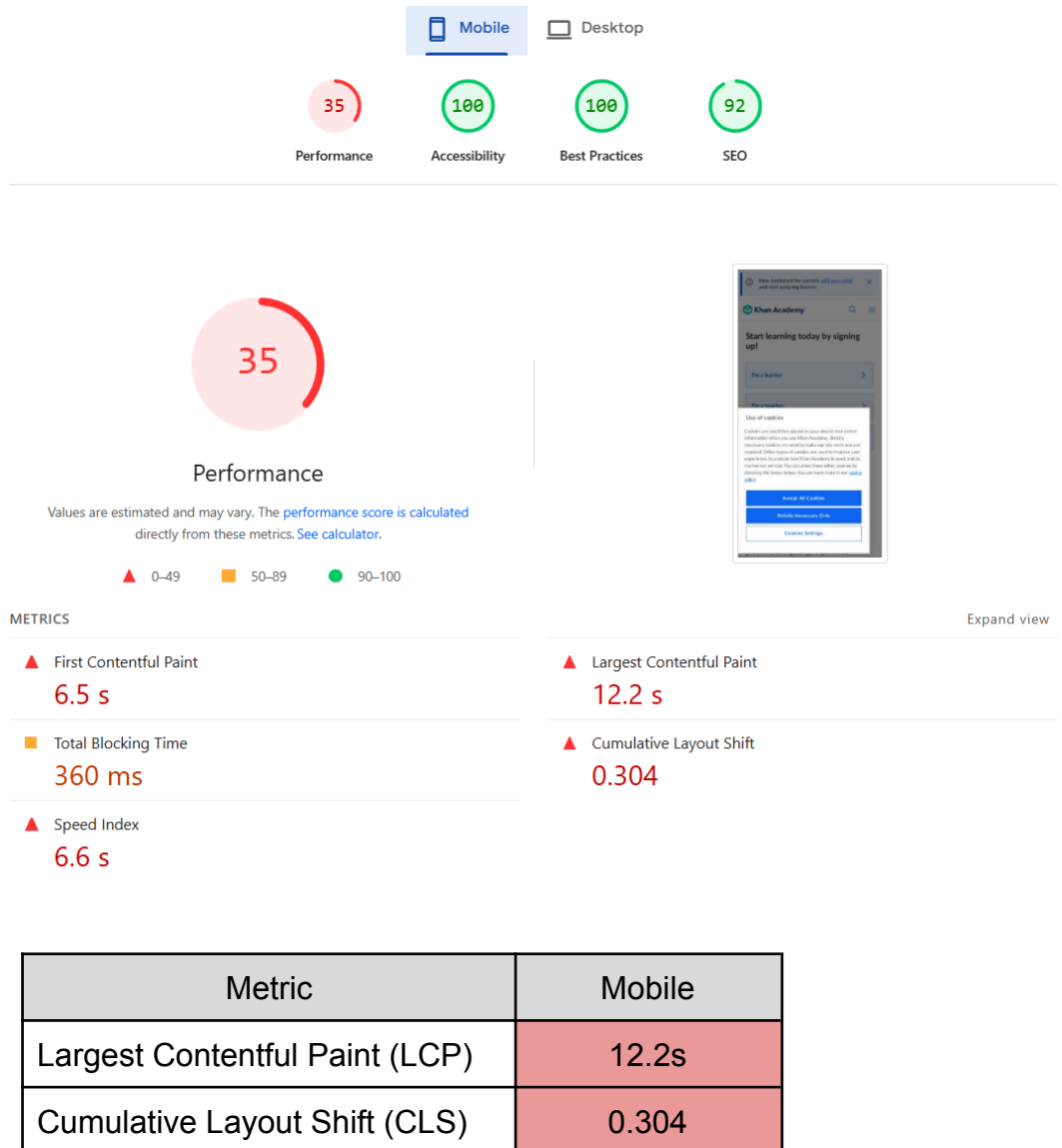
2.3.2 Mobile

Global Data:



Metric	Mobile
Largest Contentful Paint (LCP)	9.1s
Interaction to Next Paint (INP)	585ms
Cumulative Layout Shift (CLS)	0.29

Local
(Lab) Data:



2.3.3 Local (Lab) Measurements:

- Largest Contentful Paint - (LCP):

Desktop:

The LCP result is 2.5s. This indicates that the website loads the largest content element at 2.5s, which is in the 'Needs Improvement' time range. A slight improvement will be enough to make it in the 'Good' time range (optimal time - Good < 2.5s).

The Specific Element that causes LCP -

```

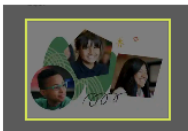
```

○ LCP breakdown

Each [subpart has specific improvement strategies](#). Ideally, most of the LCP time should be spent on loading the resources, not within delays. [LCP](#)

Unscored

Subpart	Duration
Time to first byte	0 ms
Resource load delay	1,490 ms
Resource load duration	30 ms
Element render delay	620 ms



```
div_1s6u9f56 > div_8plbkmf > div_6ui5xj3 > img_82vv51  

```

Mobile:

The LCP result is 12.2s. This indicates that the website loads the largest content element at 12.2s, which is in the 'Poor' time range. (optimal time needed - Good < 2.5s)

The Specific Element that causes LCP -

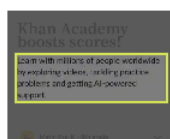
```
<span class="_12dce7ct">
```

○ LCP breakdown

Each [subpart has specific improvement strategies](#). Ideally, most of the LCP time should be spent on loading the resources, not within delays. [LCP](#)

Unscored

Subpart	Duration
Time to first byte	0 ms
Element render delay	1,910 ms



```
Learn with millions of people worldwide by exploring videos, tackling practice ...  
<span class="_12dce7ct">
```


Differences and Reasons:

Desktop LCP	Mobile LCP
2.5s - Needs Improvement	12.2s - Poor

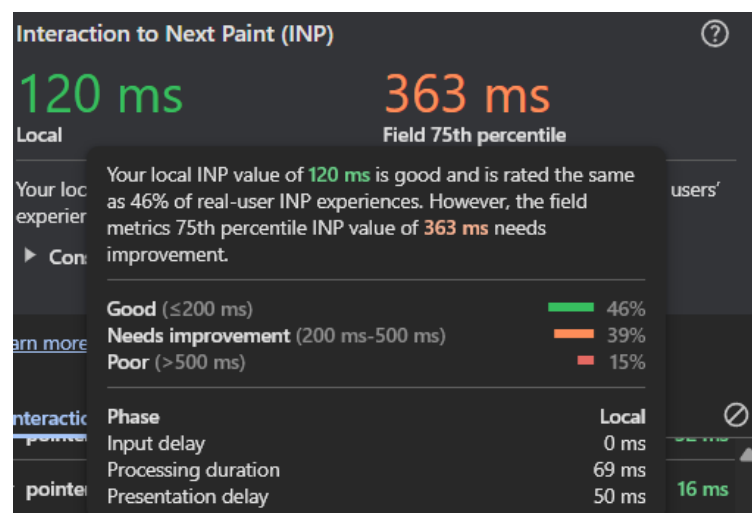
Desktop LCP (2.5s) is a lot faster than Mobile LCP (12.2s) but Desktop still needs a slight improvement to be at it's best. The LCP breakdown shows us how mobile has experienced a higher delay than it did on desktop, which is due to render-blocking requests (~ 6 seconds) and a much larger element render delay (mobile - 1,910ms, desktop - 620ms). Many other reasons may have contributed to this too. Mobiles suffering from slower CPU performance, network throttling, and longer JS executions.

- Interaction to Next Point - (INP):

Desktop:

Metric	Desktop
Interaction to Next Paint (INP)	120ms

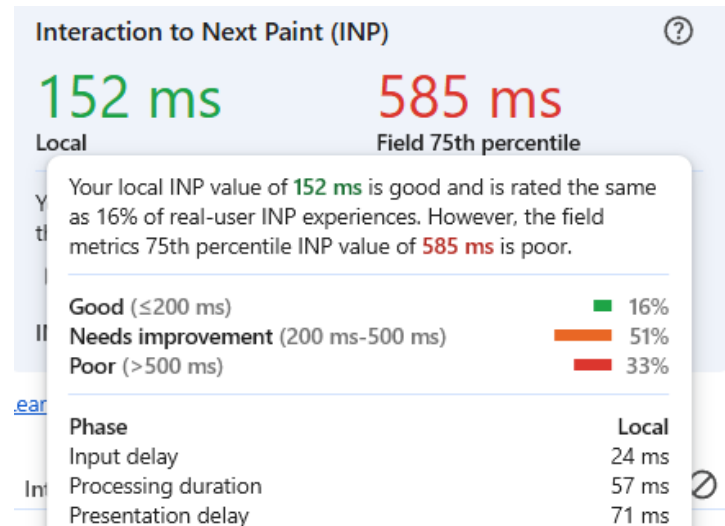
When testing, everything interacted with us instantly. Devtools results show how long each phase took. Input Delay was 0ms, Processing Duration took 69ms, and finally, presentation delay was 50ms. Totaling to ~120ms. Which is in the 'Good' time range. (optimal time - Good < 200ms)



Mobile:

Metric	Mobile
Interaction to Next Paint (INP)	152ms

When testing, everything interacted with us instantly. Devtools results show how long each phase took. Input Delay was 24ms, Processing Duration took 57ms, and finally, presentation delay was 71ms. Totaling to 152ms. Which is in the 'Good time range. (optimal time - Good < 200ms).



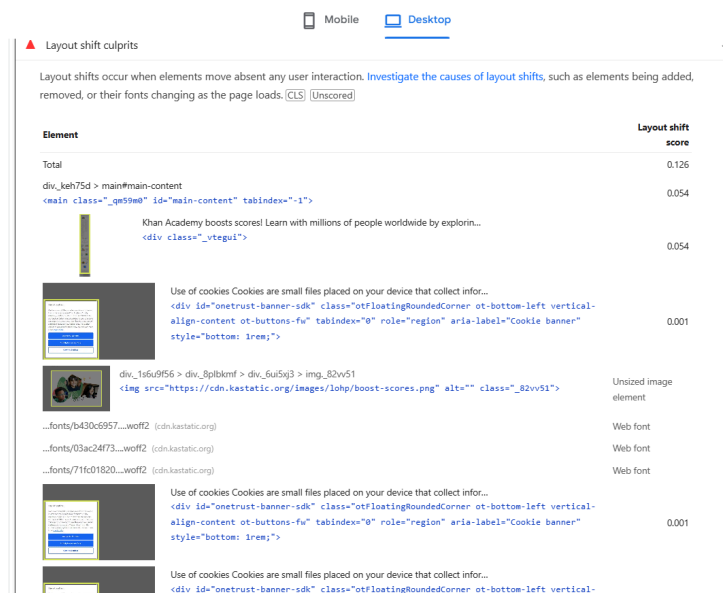
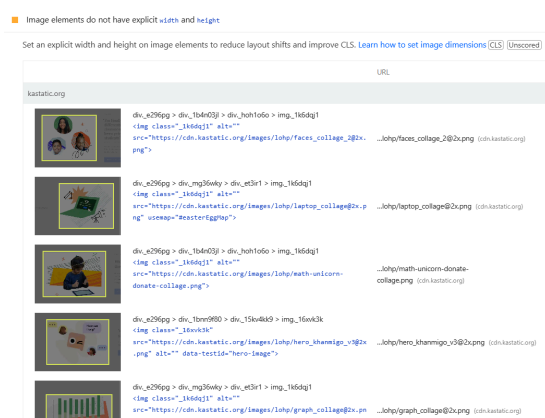
- Cumulative Layout Shift - (CLS):

Desktop:

Metric	Desktop
Cumulative Layout Shift (CLS)	0.126

The CLS result obtained is 0.126, which indicates a slight noticeable shifting when loading. It is in the 'Needs improvement range' (optimal shift score - Good < 0.1).

The main content area has created a lot of layout shifts. (0.126 and 0.054). Total - 0.126. The shifts happened by the dynamic content (elements) loading in many stages. Cookie banner has contributed very little too after the page loaded. There were also many images with no explicit width and height.



The Element that causes CLS -

```
<div id="onetrust-banner-sdk" class="otFloatingRoundedCorner
ot-bottom-left vertical-align-content ot-buttons-fw" tabindex="0"
role="region" aria-label="Cookie banner" style="bottom: 1rem;"> -
cookie
```

```

```

```
<main class="_qm59m0" id="main-content" tabindex="-1">
```

... and many more which have no explicit width & height

Mobile:

Metric	Mobile
Cumulative Layout Shift (CLS)	0.304

The CLS result obtained is 0.304, which indicates extreme shifting when loading. It is in the 'Poor time range' (optimal shift score - Good < 0.1).

The main content area has caused the largest layout shifts (0.304) which indicates that multiple elements inside has shifted during loading stages. Cookie Banner and Content Block both contributed 0.102-each to the Layout Shift.

Layout shift culprits

Layout shifts occur when elements move absent any user interaction. [Investigate the causes of layout shifts](#), such as elements being added, removed, or their fonts changing as the page loads. [CLS](#) [Unscored](#)

Element	Layout shift score
Total	0.304
div_keh75d > main#main-content	0.102
<main class="_qm59m0" id="main-content" tabindex="-1">	0.102
Khan Academy boosts scores! Learn with millions of people worldwide by explorin...	0.102
<div class="_vtegui">	0.102
Use of cookies Cookies are small files placed on your device that collect infor...	0.005
<div id="onetrust-banner-sdk" class="otFloatingRoundedCorner ot-bottom-left vertical-align-content ot-buttons-fw" tabindex="0" role="region" aria-label="Cookie banner" style="bottom: 1rem;">	0.005
...fonts/b430c6957...woff2 (cdn.kastatic.org)	Web font
...fonts/03ac24f73...woff2 (cdn.kastatic.org)	Web font
...fonts/71fc01820...woff2 (cdn.kastatic.org)	Web font
Use of cookies Cookies are small files placed on your device that collect infor...	0.003
<div id="onetrust-banner-sdk" class="otFloatingRoundedCorner ot-bottom-left vertical-align-content ot-buttons-fw" tabindex="0" role="region" aria-label="Cookie banner" style="bottom: 1rem;">	0.003
Use of cookies Cookies are small files placed on your device that collect infor...	0.002
<div id="onetrust-banner-sdk" class="otFloatingRoundedCorner ot-bottom-left vertical-align-content ot-buttons-fw" tabindex="0" role="region" aria-label="Cookie banner" style="bottom: 1rem;">	0.002

Finally, many images were noted to have no explicit width and height.

Image elements do not have explicit width and height

Set an explicit width and height on image elements to reduce layout shifts and improve CLS. [Learn how to set image dimensions](#) [CLS](#) [Unscored](#)

URL
kastatic.org
div_e296pg > div_1b4n03j > div_hoh106o > img_1k6dq1 src="https://cdn.kastatic.org/images/lohp/faces_collage_2@2x.png" ...lohp/faces_collage_2@2x.png (cdn.kastatic.org)
div_e296pg > div_mg36wky > div_et3ir1 > img_1k6dq1 src="https://cdn.kastatic.org/images/lohp/laptop_collage@2x.png" ...lohp/laptop_collage@2x.png (cdn.kastatic.org)
div_e296pg > div_1b4n03j > div_hoh106o > img_1k6dq1 src="https://cdn.kastatic.org/images/lohp/math-unicorn-donate-collage.png" ...lohp/math-unicorn-donate-collage.png (cdn.kastatic.org)
div_e296pg > div_1b4n03j > div_hoh106o > img_1k6dq1 src="https://cdn.kastatic.org/images/lohp/hero_khanmigo_v3@2x.png" alt="" data-testid="hero-image" ...lohp/hero_khanmigo_v3@2x.png (cdn.kastatic.org)
div_e296pg > div_mg36wky > div_et3ir1 > img_1k6dq1 src="https://cdn.kastatic.org/images/lohp/graph_collage@2x.png" ...lohp/graph_collage@2x.png (cdn.kastatic.org)

```
<div id="onetrust-banner-sdk" class="otFloatingRoundedCorner  
ot-bottom-left vertical-align-content ot-buttons-fw" tabindex="0"  
role="region" aria-label="Cookie banner" style="bottom: 1rem;"> -  
cookie  
  
<main class="_qm59m0" id="main-content" tabindex="-1">  
  
  
  
... and many more <img> which have no explicit width & height
```

- Mobile vs Desktop Performance Comparison Table

Metric	Mobile	Desktop
Performance Score	35	69
Largest Contentful Paint (LCP)	12.2s (Poor)	2.5s (Needs Improvement)
Interaction to Next Paint (INP)	152ms (Good)	120ms (Good)
Cumulative Layout Shift (CLS)	0.304 (Poor)	0.126 (Needs Improvement)

Overall, the desktop version of the website is much better in all metrics, reaching 'Good Ranges' in INP where no improvements are needed. The mobile version has 2 areas that needs to be fixed upon instantly (LCP and CLS), which indicates that there is a significantly long loading time for the largest visible content, and huge amount of shifting during the loading time. Although Desktop is the clear winner it needs to improve both its LCP and CLS so that they may be in the 'Good Range'. Mobile did well for INP.

Top 3 Issues Flagged by Lighthouse

Desktop:

Issue	Type	Savings
Improve image delivery	Images	1,775 kib
Render blocking requests	CSS/JS	430 ms
Reduce Unused JS	JS	515 kib

INSIGHTS	DIAGNOSTICS
▲ Render blocking requests — Est savings of 430 ms	▲ Reduce unused JavaScript — Est savings of 515 KiB
▲ Use efficient cache lifetimes — Est savings of 51 KiB	▲ Reduce JavaScript execution time — 1.3 s
▲ Font display — Est savings of 30 ms	▲ Minimize main-thread work — 2.2 s
▲ Layout shift culprits	■ Image elements do not have explicit <code>width</code> and <code>height</code>
▲ Forced reflow	■ Reduce unused CSS — Est savings of 28 KiB
▲ LCP request discovery	■ Avoid enormous network payloads — Total size was 5,475 KiB
▲ Network dependency tree	○ Avoid long main-thread tasks — 4 long tasks found
■ Improve image delivery — Est savings of 1,775 KiB	○ User Timing marks and measures — 2 user timings
■ Legacy JavaScript — Est savings of 19 KiB	○ Avoid non-composited animations — 3 animated elements found
○ Optimize DOM size	
○ LCP breakdown	

Mobile:

Issue	Type	Savings
Render blocking requests	CSS/JS	5,960 ms
Improve image delivery	Images	1,257 kib
Reduce unused JS	JS	515 kib

INSIGHTS

- ▲ Render blocking requests — Est savings of 5,960 ms
- ▲ Use efficient cache lifetimes — Est savings of 51 KiB
- ▲ Layout shift culprits
- ▲ Forced reflow
- ▲ Network dependency tree
- Font display — Est savings of 10 ms
- Improve image delivery — Est savings of 1,257 KiB
- Legacy JavaScript — Est savings of 19 KiB
- LCP breakdown
- 3rd parties

DIAGNOSTICS

- ▲ Reduce JavaScript execution time — 1.8 s
- ▲ Minimize main-thread work — 2.7 s
- ▲ Reduce unused JavaScript — Est savings of 515 KiB
- Image elements do not have explicit `width` and `height`
- Reduce unused CSS — Est savings of 27 KiB
- Avoid enormous network payloads — Total size was 4,671 KiB
- Avoid long main-thread tasks — 5 long tasks found
- User Timing marks and measures — 2 user timings
- Avoid non-composited animations — 1 animated element found

3. Comparative Analysis & Optimization Strategy

Comparison Table:

Category	Nike (E-commerce)	Al-Jazeera (News/Media)	KhanAcademy (Education)
Overall Performance (0-100)	Desktop: 93 Mobile: 62	Desktop: 63 Mobile: 53	Desktop: 69 Mobile: 35
Average Load Time	4.15s	16.32s	4.28s
Average Page Size (Transferred)	2.2 MB	4.4 MB	4 MB
Average Requests	142	241	136
Core Web Vitals (Desktop)	LCP: 1.2s (Good) INP: 120ms (Good) CLS: 0.041 (Good)	LCP: 0.9s (Good) INP: 144ms (Good) CLS: 0.059 (Good)	LCP: 2.5s (Needs Improvement) INP: 120ms (Good) CLS: 0.126 (Needs Improvement)
Core Web Vitals (Mobile)	LCP: 3.5s (Needs Improvement) INP: 184ms (Good) CLS: 0.182 (Needs Improvement)	LCP: 4.6s (Poor) INP: 432ms (Needs Improvement) CLS: 0.0 (Good)	LCP:12.2s (Poor) INP: 152ms (Good) CLS: 0.304 (Poor)
Performance Ranking	#1	#2	#3

Why?

#1 (Nike):

- It has the highest desktop score - 93
- A good mobile score - 62
- It's Overall Core Web Vital Performance are all in the 'Good' Range

#2 (Al-Jazeera):

- Medium desktop score - 63
- Medium mobile score - 53
- Its Core Web Vital performance is mixed. Good on desktop overall but on mobile it has 'poor' LCP and 'needs improvement' INP
- Heavier number of requests but still better than KhanAcademy in the overall

#3 (KhanAcademy):

- Reasonable desktop score - 69
- Very low mobile score - 35
- Core Web Vital Performance shows how LCP and CLS does very bad on both mobile and desktop (its Desktop LCP is 0.1 away from being good - it needs to be improved slightly)
- Very poor mobile LCP - 12.2s

Worst-Performing Website (KhanAcademy):

A. Identifying Performance Bottleneck (Mobile)

1. Render-Blocking Resources (CSS/JS):

It was flagged as a 'Render blocking requests' with an estimated savings of 5,960 ms. These important CSS&JS are etched before the page can even render it, which then affects LCP greatly.

2. Excessive JS Execution:

It was flagged as a 'Reduce JS execution time' - 1.4s, 'Reduce unused JS' ~515 kib.

3. Unoptimized Image Delivery:

It was flagged as a 'Improve image delivery' ~1,257. The LCP element and the other hero-images are very heavy, unoptimized, and have no explicit width or height. All of which contributes to slow LCP and increase layout shifts (Note: CLS - 0.304)

4. Inefficient Caching/Resource Loading:

It was flagged as a 'Use efficient cache lifetimes', which means there is waste of time and missed optimizations due to short Cache TTL (Time to Live-1 Day). Meaning? When the same users keep revisiting, they have to re-download the resources even though they might already have it.

B. Concrete Optimization Recommendations:

1. Rendering Paths Optimization:

- Place the essential CSS stylesheets you want the users to see first directly in the HTML. This way, you will not have any delays during which the browser needs to locate and apply the CSS file, since there will be no 'stop and wait' delays for when the browser needs to find and use the css file.
- Use fetchpriority="high" on LCP image
- Add a preconnect to the CDN to create a connection early, which would save time and help reduce LCP significantly.

2. Image Optimization:

- Transform hero images into Webp or AVIF format.
- Include explicit width/height for all images to prevent layout shifts.
- Add lazy loading to all the images that are off screen to prioritize the main and critical content. As soon as the user gets near to the image that is off screen it then gets loaded.

3. JavaScript Delivery:

- Implement a route based code splitting for React Application:
- Remove all of the unused modules
- Modernize the build target so that it can reduce legacy ~19 kib.
- Defer any third-party scripts until after initial rendering.

4. Caching Strategy:

- Extend the cache TTL to a more suitable amount.
- Use Service Worker for offline capabilities and performance on repeat visits.

5. Layout Stability Improvements:

- Reserve the space for cookie banner before it may even load
- Add CSS containment to the major content sections.

Conclusion

This report has demonstrated how each website's performance varies significantly across the different domains when evaluating it using the structured 3-phase methodology. By recording and collecting technology stacks, Global and Local performance measurements, and detailed Core Web Vitals comparisons, we identified each website's strengths and weaknesses. Nike has performed the best overall, Al-Jazeera showed a mixed of results, but they were stable, and KhanAcademy revealed major bottlenecks, especially in mobile LCP and CLS. The findings showed and highlighted how important it is to optimize continuously, especially for content heavy websites, to ensure a fast, stable, and user friendly experience across all devices.